

**ULTIMATE SOLUTION BOOK:
ALL EXERCISES & PAST EXAMS**

Mathematics for Business B

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

2027 EDITION

LET ME INTRODUCE MYSELF.

About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- ✓ My name is **Jovan Samardzic**, I am 26 years old
- ✓ Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- ✓ Recipient of the **Science@Leuven Scholarship**

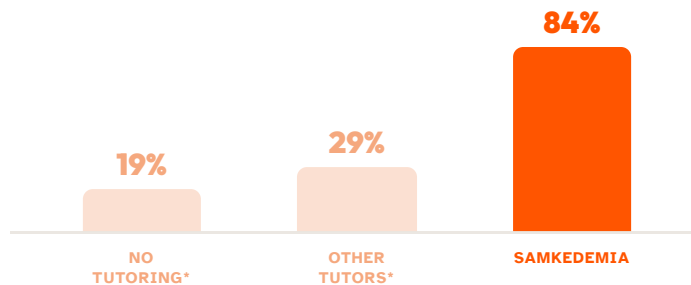
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- ✓ Founder of **Samkedia**, helping students pass their exams
- ✓ Worked with **200+ university students** from all kinds of backgrounds
- ✓ **Plain language, not textbook theory** - just what actually makes sense

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Contents

1	Chapter 1 - Functions of Two Variables	5
	Chapter 1 - Question 1	5
	Chapter 1 - Question 2	16
	Chapter 1 - Question 3	29
	Chapter 1 - Question 4	34
	Chapter 1 - Question 5	37
	Chapter 1 - Question 6	39
	Chapter 1 - Question 7	41
	Chapter 1 - Question 8	50
	Chapter 1 - Question 9	52
	Chapter 1 - Question 10	54
	Chapter 1 - Question 11	57
	Chapter 1 - Question 12	59
	Chapter 1 - Question 13	64
	Chapter 1 - Question 14	69
	Chapter 1 - Question 15	72
	Chapter 1 - Question 16	78
	Chapter 1 - Question 17	84
	Chapter 1 - Question 18	88
	Chapter 1 - Question 19	98
2	Review 1 - Functions of Two Variables	103
	Review 1 - Question 1	103
	Review 1 - Question 2	109
	Review 1 - Question 3	114
	Review 1 - Question 4	119
3	Chapter 2 - Linear functions and linear programming	124

4	Review 2 - Linear functions and linear programming	125
5	Chapter 3 - Partial Derivatives	126
6	Review 3 - Partial Derivatives	127
7	Chapter 4 - Optimization	128
8	Review 4 - Optimization	129
9	Chapter 5 - Sequences and Series	130
10	Review 5 - Sequences and Series	131
11	Exam: 2020 September Online	132
12	Exam: 2020 September On-Campus	133
13	Exam: 2021 Example	134
14	Exam: 2021 June	135
15	Exam: 2022 Example	136
16	Exam: 2025 Example	137
17	Exam: 2025 June	138

Chapter 1 - Functions of Two Variables

Question 1

Find the domains of the following functions of two variables.

(a) $f(x, y) = \frac{xy}{\sqrt{x^2 - 1}}$

(b) $f(x, y) = \sqrt{(x - 1)(2 + y)}$

(c) $f(x, y) = \ln(4 - x^2 - y^2)$

(d) $f(x, y) = \frac{1}{x + y - 2}$

(e) $f(x, y) = \sqrt{y - x + 1}$

(f) $f(x, y) = \frac{1}{\sqrt{xy}}$

(g) $f(x, y) = \log_2(x^2 + y^2 - 9)$

Tip: Understanding Constraints

- Whenever we have a square root $\sqrt{\text{something}}$, the thing inside must be **zero or positive** (never negative), otherwise, the function is not defined.
- A denominator must **never be zero**, because division by zero is not allowed.

💡 Tip: Understanding Vertical and Horizontal Lines

- The equation $x = a$ represents a **vertical line** at $x = a$ because every point on this line has the same x -value while y can be anything.
- The equation $y = a$ represents a **horizontal line** at $y = a$ because every point on this line has the same y -value while x can be anything.

✓ Solution 1: (a) Finding the domain of $f(x, y) = \frac{xy}{\sqrt{x^2-1}}$

Step 1: Identify constraints

The denominator contains a square root, so we impose the conditions:

$$x^2 - 1 \geq 0$$

$$x^2 - 1 \neq 0$$

Step 2: Solve the first constraint using a sign chart

Rewriting the inequality:

$$x^2 - 1 \geq 0$$

Factoring:

$$(x - 1)(x + 1) \geq 0$$

Solving using a sign chart:

x		-1		1	
$(x - 1)$	$-$	$-$	$-$	0	$+$
$(x + 1)$	$-$	0	$+$	$+$	$+$
$(x - 1)(x + 1)$	$+$	0	$-$	0	$+$

From the sign chart, $(x - 1)(x + 1) \geq 0$ means:

$$x \leq -1 \quad \text{or} \quad x \geq 1$$

Step 3: Solve the second constraint

$$x^2 - 1 \neq 0$$

This means:

$$(x - 1)(x + 1) \neq 0$$

So we must exclude $x = \pm 1$.

Step 4: Combine the two conditions

- From Step 2: $x \leq -1$ or $x \geq 1$.
- From Step 3: $x \neq -1$ and $x \neq 1$.

Since both conditions must be true at the same time, we remove $x = \pm 1$, giving:

$$x < -1 \quad \text{or} \quad x > 1$$

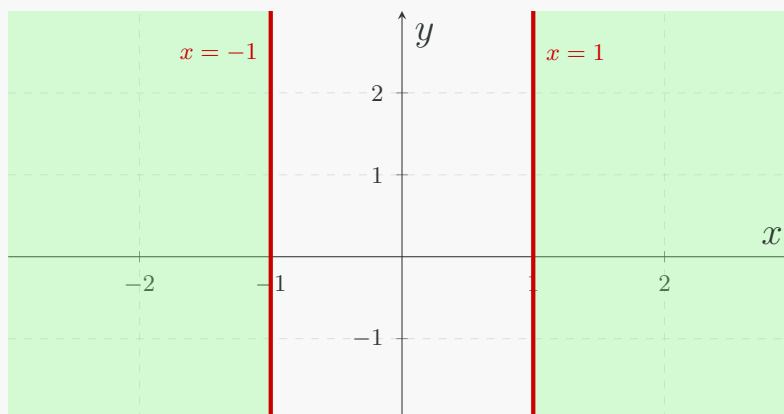
Since there are no restrictions on y , it can be any real number.

Final Domain:

$$\{(x, y) \in \mathbb{R}^2 \mid x < -1 \text{ or } x > 1, \quad y \in \mathbb{R}\}$$

Graphical Representation:

The valid domain is highlighted in green, while the excluded points ($x = \pm 1$) are marked with red lines.



The vertical red lines at $x = 1$ and $x = -1$ indicate that these values are excluded from the domain.

Answer: The domain is $\{(x, y) \in \mathbb{R}^2 \mid x < -1 \text{ or } x > 1, y \in \mathbb{R}\}$.

✓ **Solution 1: (b) Finding the domain of $f(x, y) = \sqrt{(x - 1)(2 + y)}$**

Step 1: Identify constraints

Since the expression under the square root must be non-negative:

$$(x - 1)(2 + y) \geq 0$$

Step 2: Solve the inequality by analyzing cases

The inequality we need to solve is:

$$(x - 1)(2 + y) \geq 0$$

Since this is a product of two factors, it is non-negative when:

- Both factors are positive:

$$x - 1 \geq 0 \quad \text{and} \quad 2 + y \geq 0 \quad \Rightarrow \quad x \geq 1 \quad \text{and} \quad y \geq -2.$$

- Both factors are negative:

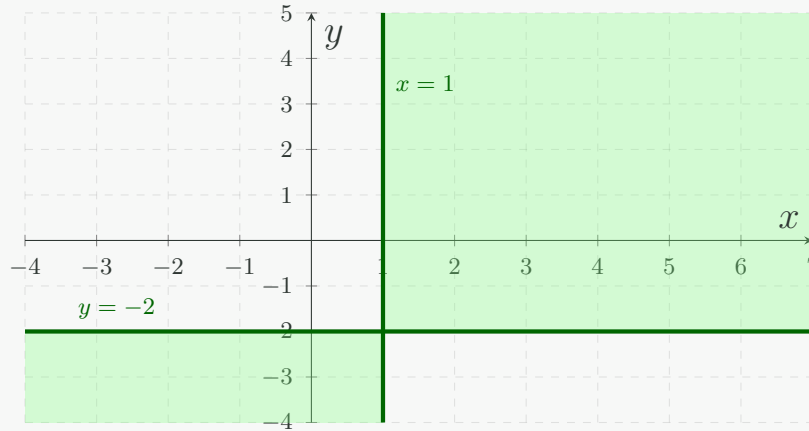
$$x - 1 \leq 0 \quad \text{and} \quad 2 + y \leq 0 \quad \Rightarrow \quad x \leq 1 \quad \text{and} \quad y \leq -2.$$

Final Domain:

$$\{(x, y) \in \mathbb{R}^2 \mid (x \geq 1, y \geq -2) \text{ or } (x \leq 1, y \leq -2)\}$$

Graphical Representation:

The valid domain is highlighted in green.



Answer: The domain is $\{(x, y) \in \mathbb{R}^2 \mid (x \geq 1, y \geq -2) \text{ or } (x \leq 1, y \leq -2)\}$.

💡 Tip: Understanding Logarithm Constraints

The expression inside a logarithm $\ln(\text{something})$ must be **strictly positive** (greater than zero). It cannot be zero or negative.

💡 Tip: Understanding Circle Equations

- The equation $x^2 + y^2 = r^2$ represents a **circle** centered at $(0, 0)$ with radius r .
- If the inequality is $x^2 + y^2 < r^2$, it represents the **inside** of the circle.
- If the inequality is $x^2 + y^2 > r^2$, it represents the **outside** of the circle.

✓ Solution 1: (c) Finding the domain of $f(x, y) = \ln(4 - x^2 - y^2)$ **Step 1: Identify constraints**

Since the logarithm must have a positive argument:

$$4 - x^2 - y^2 > 0$$

Step 2: Solve the inequality

Rearrange:

$$4 > x^2 + y^2$$

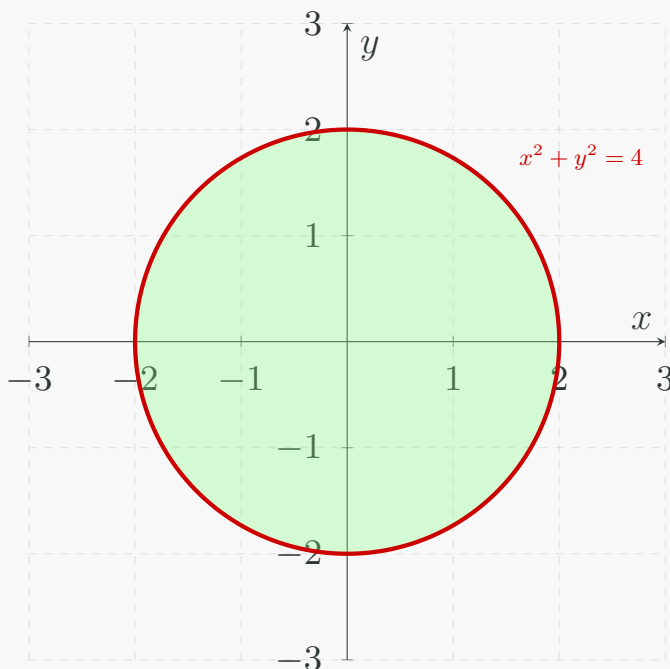
Hence, this represents a circle centered at $(0, 0)$ with radius 2, but the inequality is strict ($<$), meaning the points on the circle's edge are **not included** in the domain.

Final Domain:

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 4\}$$

Graphical Representation:

The valid domain is highlighted in green, while the excluded circle boundary is marked in red.



Answer: The domain is $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < 4\}$.

 Tip: Understanding Denominator Constraints

A denominator must **never be zero**, because division by zero is undefined.

 Tip: Understanding Linear Equations

An equation of the form $y = mx + b$ represents a **straight line** with slope m and y-intercept b .

 Solution 1: (d) Finding the domain of $f(x, y) = \frac{1}{x+y-2}$ **Step 1: Identify constraints**

Since the denominator cannot be zero, we impose the condition:

$$x + y - 2 \neq 0$$

Step 2: Solve for the excluded values

Rearrange the equation:

$$y \neq 2 - x$$

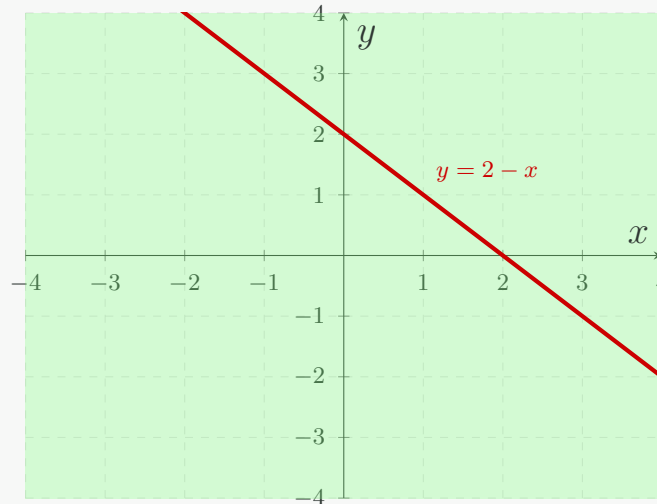
Hence, this equation represents a straight line $y = 2 - x$. Every point on this line is excluded from the domain.

Final Domain:

$$\{(x, y) \in \mathbb{R}^2 \mid y \neq 2 - x\}$$

Graphical Representation:

The valid domain is highlighted in green, while the excluded line $y = 2 - x$ is marked in red.



Answer: The domain is $\{(x, y) \in \mathbb{R}^2 \mid y \neq 2 - x\}$.

💡 Tip: Understanding Linear Inequalities

- An equation of the form $y = mx + b$ represents a **straight line** with slope m and y-intercept b .
- When the inequality is $y \geq mx + b$, the valid region is **above or on the line**.
- When the inequality is $y \leq mx + b$, the valid region is **below or on the line**.

✔ Solution 1: (e) Finding the domain of $f(x, y) = \sqrt{y - x + 1}$

Step 1: Identify constraints

Since the square root must have a non-negative argument:

$$y - x + 1 \geq 0$$

Step 2: Solve for the valid region

Rearrange the inequality:

$$y \geq x - 1$$

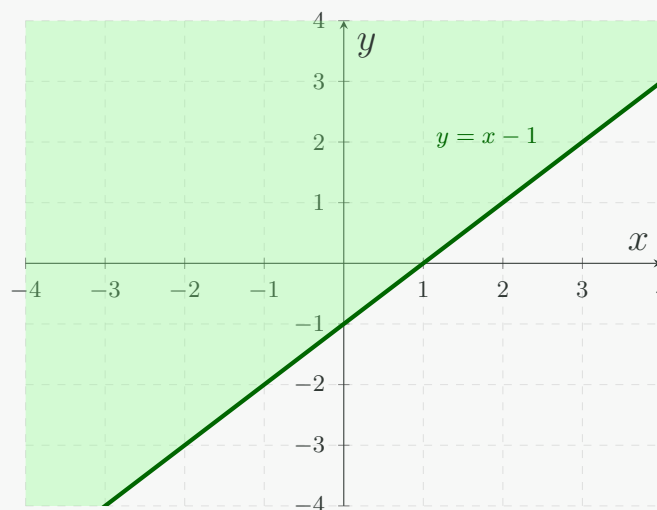
Hence, this represents the region **above or on** the line $y = x - 1$, since the inequality includes $\geq$, meaning points on the line itself are included in the domain.

Final Domain:

$$\{(x, y) \in \mathbb{R}^2 \mid y \geq x - 1\}$$

Graphical Representation:

The valid domain is highlighted in green.



Answer: The domain is $\{(x, y) \in \mathbb{R}^2 \mid y \geq x - 1\}$.

✓ Solution 1: (f) Finding the domain of $f(x, y) = \frac{1}{\sqrt{xy}}$

Step 1: Identify constraints

Since we have both a square root and a denominator, we must check two conditions separately:

- The expression inside the square root must be non-negative:

$$xy \geq 0$$

- The denominator cannot be zero:

$$xy \neq 0$$

Step 2: Solve for the valid regions

The condition $xy \geq 0$ means x and y must have the same sign:

- If $x > 0$, then $y \geq 0$ (both positive or zero).
- If $x < 0$, then $y \leq 0$ (both negative or zero).

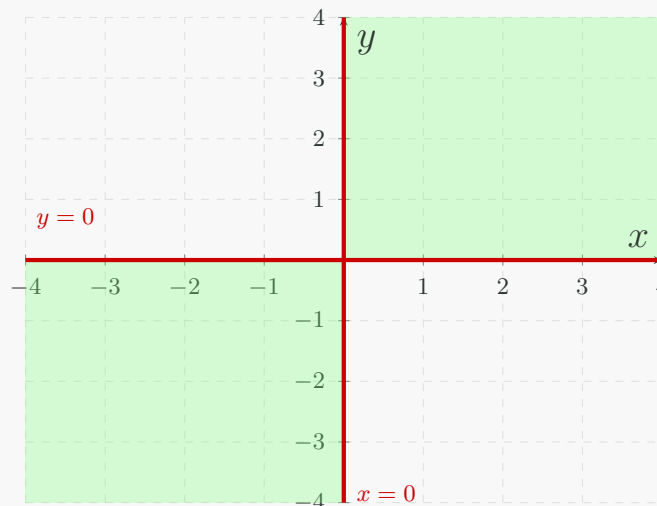
However, the additional condition $xy \neq 0$ means that we must exclude the x- and y-axes ($x = 0$ and $y = 0$), since at least one variable being zero would make $xy = 0$.

Final Domain:

$$\{(x, y) \in \mathbb{R}^2 \mid xy > 0\}$$

This means the function is defined in the first and third quadrants, but the x- and y-axes are excluded.

Graphical Representation: The valid domain is highlighted in green, while the excluded x- and y-axes are marked in red.



Answer: The domain is $\{(x, y) \in \mathbb{R}^2 \mid xy > 0\}$

✓ Solution 1: (g) Finding the domain of $f(x, y) = \log_2(x^2 + y^2 - 9)$ **Step 1: Identify constraints**

Since the logarithm must have a positive argument:

$$x^2 + y^2 - 9 > 0$$

Step 2: Solve the inequality

Rearrange:

$$x^2 + y^2 > 9$$

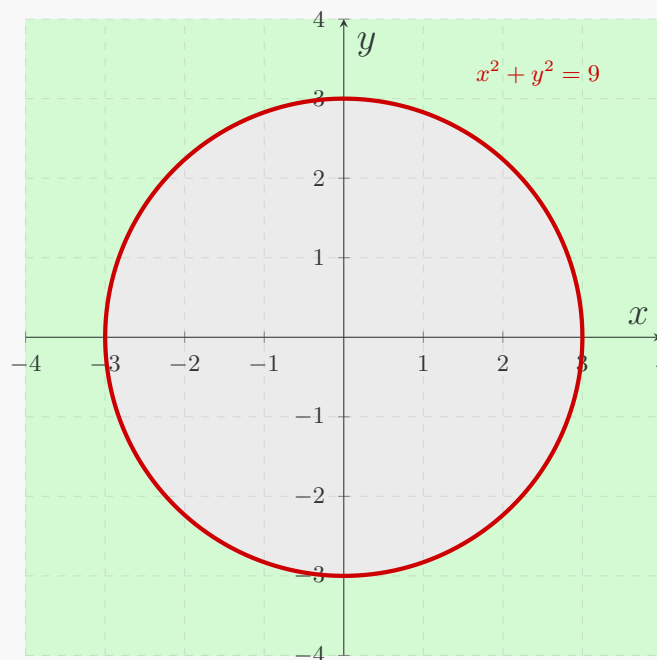
This represents the outside of a circle centered at $(0, 0)$ with radius 3. Since the inequality is strict ($>$), the points on the circle itself are not included in the domain.

Final Domain:

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 > 9\}$$

Graphical Representation:

The valid domain is highlighted in green, while the excluded circle boundary is marked in red.



Answer: The domain is $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 > 9\}$.

Question 2

Make the sign chart for the following continuous functions of two variables.

(a) $f(x, y) = 2x - 3y + 6$

(b) $f(x, y) = (x + 1)^2 + y^2 - 1$

(c) $f(x, y) = x^2 - xy + x$

(d) $f(x, y) = x^2 + y^2 - x$

(e) $f(x, y) = \frac{\sqrt{x - y}}{y}$

(f) $f(x, y) = \frac{\ln(x - y)}{y}$

Tip: Key Steps for Sign Charts

To determine where a function of two variables is positive or negative, follow these three steps:

1. **Find the domain** – Check for any restrictions on x and y .
2. **Find where the function is zero** – Solve $f(x, y) = 0$ to identify zeroes. These are going to serve as boundary lines or curves.
3. **Find the sign in each region** – Pick a test point in each region to determine whether the function is positive or negative.

💡 Tip: Understanding Sign Charts for Linear Functions

For a function of the form $f(x, y) = ax + by + c$:

- The function is **zero** on the line $ax + by + c = 0$.
- The function is **positive** on one side of the line and **negative** on the other.
- To determine which side is positive or negative, pick a test point (e.g., $(0, 0)$) and check the sign of $f(x, y)$.

✓ Solution 2: (a) Finding the sign chart of $f(x, y) = 2x - 3y + 6$

Step 1: Find the domain

The given function is a linear equation, which is continuous and defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Find where the function is zero

Setting $f(x, y) = 0$:

$$2x - 3y + 6 = 0$$

Rearrange for y :

$$y = \frac{2}{3}x + 2$$

This is a straight line that separates the plane into two regions.

Step 3: Identify the sign in each region

We select a test point to determine the sign of $f(x, y)$ in different regions.

- Test Point 1: $(0, 0)$

$$f(0, 0) = 2 \cdot (0) - 3 \cdot (0) + 6 = 6 > 0$$

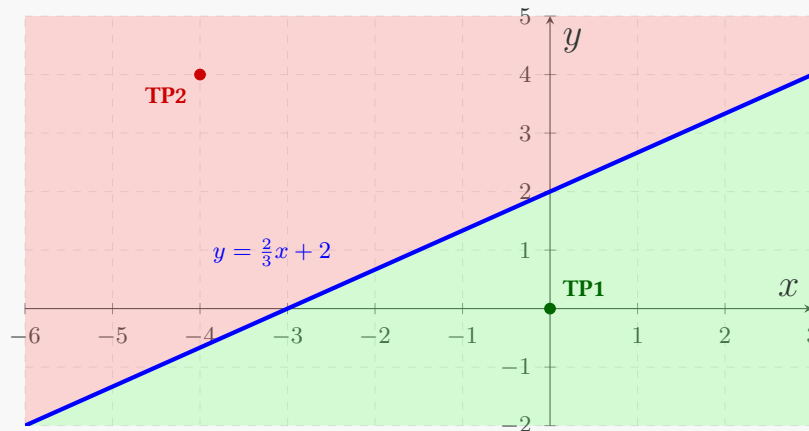
Hence, the region containing $(0, 0)$ is positive (green area).

- Test Point 2: $(-4, 4)$ (a point in the opposite region)

$$f(-4, 4) = 2 \cdot (-4) - 3 \cdot (4) + 6 = -14 < 0$$

Hence, the opposite region is negative (red area).

Answer: The green region represents where the function is positive, the red region where it is negative, and the blue line where it is zero.



💡 Tip: Understanding Sign Charts for Quadratic Functions

For a function of the form $f(x, y) = (\text{expression})^2 + (\text{expression})^2 - c$:

- The function is **zero** on the boundary (circle) equation.
- The function is **positive** outside the boundary and **negative** inside.
- To verify, pick test points inside and outside the boundary.

✓ Solution 2: (b) Finding the sign chart of $f(x, y) = (x + 1)^2 + y^2 - 1$ **Step 1: Find the domain**

The given function is a quadratic expression in x and y , and it is defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Find where the function is zero

Setting $f(x, y) = 0$:

$$(x + 1)^2 + y^2 - 1 = 0$$

Rearrange:

$$(x + 1)^2 + y^2 = 1$$

This represents a circle centered at $(-1, 0)$ with radius 1.

Step 3: Identify the sign in each region

We select a test point to determine the sign of $f(x, y)$ in different regions.

- Test Point 1: $(-1, 0)$ (center of the circle)

$$f(-1, 0) = (-1 + 1)^2 + 0^2 - 1 = -1 < 0$$

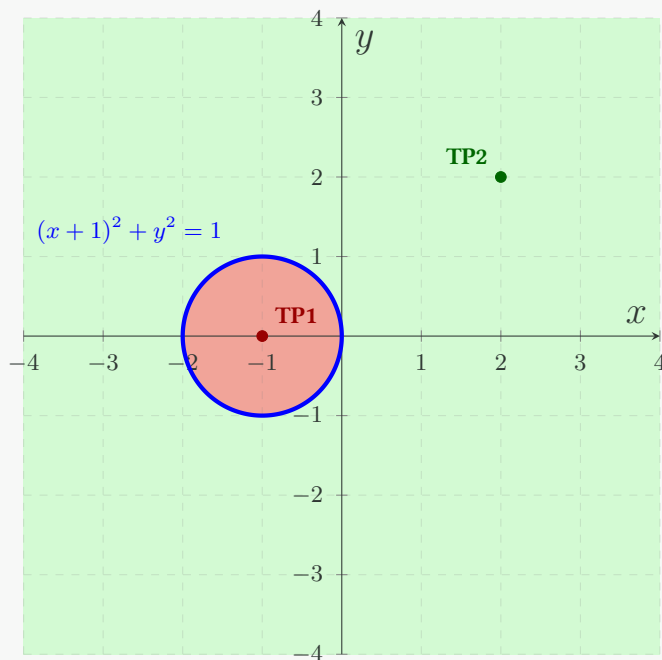
Hence, the region inside the circle is negative (red area).

- Test Point 2: $(2, 2)$ (a point outside the circle)

$$f(2, 2) = (2 + 1)^2 + 2^2 - 1 = 12$$

Hence, the region outside the circle is positive (green area).

Answer: The green region represents where the function is positive, the red region where it is negative, and the blue circle where it is zero.



✓ Solution 2: (c) Finding the sign chart of $f(x, y) = x^2 - xy + x$

Step 1: Find the domain

The given function is a quadratic polynomial in x and y , which is defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Find where the function is zero

Setting $f(x, y) = 0$:

$$x^2 - xy + x = 0$$

Factorizing:

$$x(x - y + 1) = 0$$

This gives two boundary lines:

$$x = 0 \quad \text{or} \quad y = x + 1$$

These lines separate the plane into different regions.

Step 3: Identify the sign in each region

We select test points to determine the sign of $f(x, y)$ in different regions.

- Test Point 1: $(1, 0)$

$$f(1, 0) = 1 - 0 + 1 = 2 > 0$$

Hence, the region containing $(1, 0)$ is positive (green area).

- Test Point 2: $(-1, 2)$

$$f(-1, 2) = 1 + 2 - 1 = 2 > 0$$

Hence, this region is also positive (green area).

- Test Point 3: $(-2, -2)$

$$f(-2, -2) = 4 - 4 - 2 = -2 < 0$$

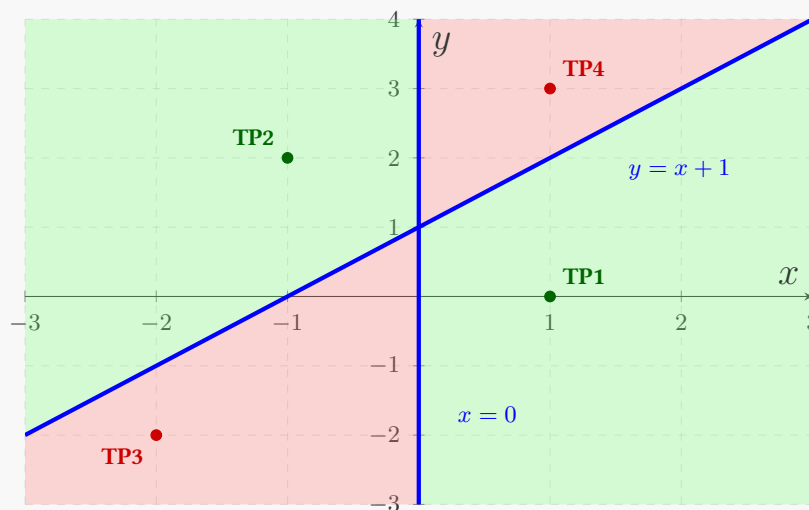
Hence, this region is negative (red area).

- Test Point 4: $(1, 3)$

$$f(1, 3) = 1 - 3 + 1 = -1 < 0$$

Hence, this region is also negative (red area).

Answer: The green region represents where the function is positive, the red region where it is negative, and the blue lines where it is zero.



Tip: Rewriting Equations into Standard Circle Form

An equation of the form:

$$x^2 + y^2 + mx + ny + l = 0$$

can be rewritten into the standard circle equation:

$$(x - a)^2 + (y - b)^2 = r^2$$

where (a, b) is the center and r is the radius. To do this:

1. Group the x - and y -terms separately.
2. Complete the square on the variable that has a linear term.
3. Rewrite in standard circle form.

This helps us identify the center and radius of the circle.

Solution 2: (d) Finding the sign chart of $f(x, y) = x^2 + y^2 - x$

Step 1: Find the domain

The function is a polynomial in x and y , which means it is defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Find where the function is zero

Setting $f(x, y) = 0$:

$$x^2 + y^2 - x = 0$$

We want to rewrite this equation in the standard form of a circle:

$$(x - a)^2 + (y - b)^2 = r^2$$

Since there is no linear y -term, we only need to complete the square for x . We rewrite:

$$x^2 - x + y^2 = 0$$

To complete the square for $x^2 - x$, we take half the coefficient of x , square it, and add/subtract:

$$x^2 - x = x^2 - 2 \cdot \frac{1}{2} \cdot x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = \left(x - \frac{1}{2}\right)^2 - \frac{1}{4}$$

Substituting this into the equation:

$$\left(x - \frac{1}{2}\right)^2 - \frac{1}{4} + y^2 = 0 \quad \Rightarrow \quad \left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$$

Thus, the function is zero along a circle centered at $\left(\frac{1}{2}, 0\right)$ with radius $\frac{1}{2}$.

Step 3: Identify the sign in each region

We select test points to determine the sign of $f(x, y)$ in different regions.

- Test Point 1: $\left(\frac{1}{2}, 0\right)$ (center of the circle)

$$f\left(\frac{1}{2}, 0\right) = \left(\frac{1}{2}\right)^2 + 0^2 - \frac{1}{2} = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4} < 0$$

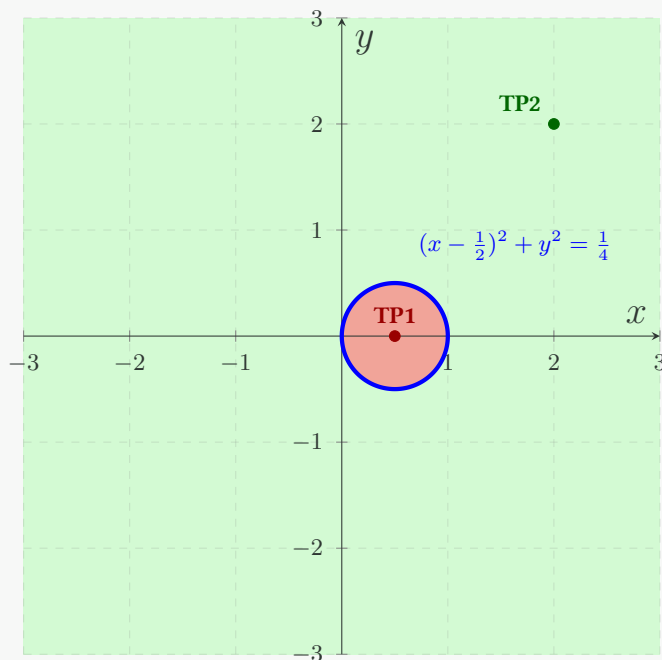
Hence, the region inside the circle is negative (red area).

- Test Point 2: $(2, 2)$ (outside the circle)

$$f(2, 2) = 2^2 + 2^2 - 2 = 4 + 4 - 2 = 6 > 0$$

Hence, the region outside the circle is positive (green area).

Answer: The green region represents where the function is positive, the red region where it is negative, and the blue circle where it is zero.



💡 Tip: Understanding Square Root and Denominator Constraints

For a function of the form $f(x, y) = \frac{\sqrt{\text{something}}}{\text{denominator}}$:

- The expression inside the square root must be non-negative.
- The denominator cannot be zero.

Both conditions must hold at the same time.

✅ Solution 2: (e) Finding the sign chart of $f(x, y) = \frac{\sqrt{x-y}}{y}$

Step 1: Find the domain

Since the function involves both a square root and a denominator, we must analyze both constraints separately.

- The square root constraint:

$$x - y \geq 0 \Rightarrow x \geq y$$

- The denominator constraint:

$$y \neq 0$$

Since both conditions must be satisfied simultaneously, we conclude:

$$\text{Domain: } \{(x, y) \in \mathbb{R}^2 \mid x \geq y, \quad y \neq 0\}$$

Step 2: Find where the function is zero

The function is zero when:

$$\sqrt{x - y} = 0 \Rightarrow x - y = 0 \Rightarrow x = y$$

This means the function is zero along the line $x = y$.

Step 3: Identify the sign in each region

We test points from two valid regions to determine the sign of $f(x, y)$.

- Test Point 1: $(2, 1)$ (above the zero line)

$$f(2, 1) = \frac{\sqrt{2-1}}{1} = 1 > 0$$

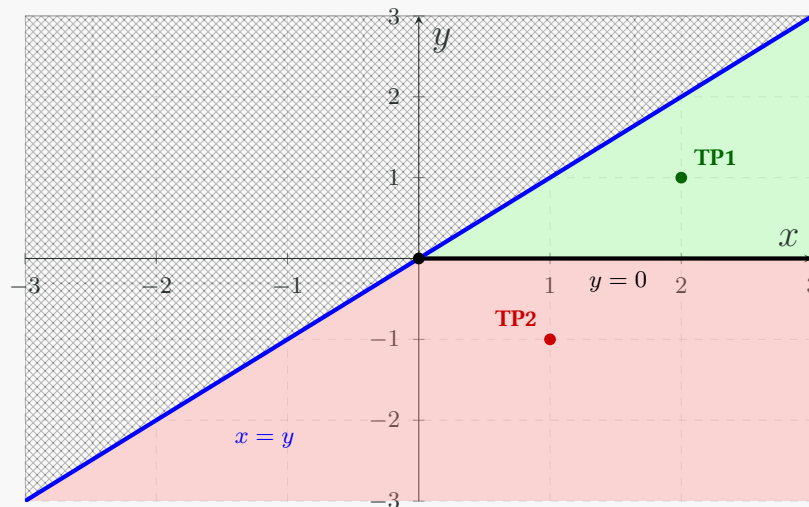
Hence, this region is positive (green area).

- Test Point 2: $(1, -1)$ (below the zero line)

$$f(1, -1) = \frac{\sqrt{1-(-1)}}{-1} = \frac{\sqrt{2}}{-1} < 0$$

Hence, this region is negative (red area).

Answer: The green region represents where the function is positive, the red region where it is negative, the blue line where it is zero, and the black-shaded region (including the x-axis) shows where the function is not defined.



💡 Tip: Understanding Logarithm and Denominator Constraints

For a function of the form $f(x, y) = \frac{\ln(\text{something})}{\text{denominator}}$:

- The argument of the logarithm must be **strictly positive**.
- The denominator cannot be zero.

Both conditions must be satisfied at the same time.

✓ Solution 2: (f) Finding the sign chart of $f(x, y) = \frac{\ln(x-y)}{y}$

Step 1: Find the domain

Since the function involves a logarithm and a denominator, we must analyze both constraints separately.

- The logarithm constraint:

$$x - y > 0 \quad \Rightarrow \quad x > y$$

- The denominator constraint:

$$y \neq 0$$

Since both conditions must hold simultaneously, we conclude:

$$\text{Domain: } \{(x, y) \in \mathbb{R}^2 \mid x > y, \quad y \neq 0\}$$

Step 2: Find where the function is zero

The function is zero when:

$$\ln(x - y) = 0 \quad \Rightarrow \quad x - y = 1 \quad \Rightarrow \quad y = x - 1$$

Thus, the function is zero along the line $y = x - 1$.

Step 3: Identify the sign in each region

We select test points from the four valid regions to determine the function's sign.

- Test Point 1: $(-1, -1.5)$ (bottom-left region)

$$f(-1, -1.5) = \frac{\ln(-1 - (-1.5))}{-1.5} = \frac{\ln 0.5}{-1.5} > 0$$

Hence, this region is positive (green area).

- Test Point 2: $(1, -1)$ (bottom-right region)

$$f(1, -1) = \frac{\ln(1 - (-1))}{-1} = \frac{\ln 2}{-1} < 0$$

Hence, this region is negative (red area).

- Test Point 3: $(3, 1)$ (top-right region)

$$f(3, 1) = \frac{\ln(3 - 1)}{1} = \ln 2 > 0$$

Hence, this region is positive (green area).

- Test Point 4: $(3, 2.5)$ (topmost region)

$$f(3, 2.5) = \frac{\ln(3 - 2.5)}{2.5} = \frac{\ln 0.5}{2.5} < 0$$

Hence, this region is negative (red area).

? Question 3

Solve the following inequalities.

(a) $2x + 6 \leq 3y$

(b) $y > y \cdot \log_2 x$

💡 Tip: Solving Inequalities in Two Variables

To solve an inequality of the form $f(x, y) \leq 0$, follow these steps:

1. **Find the domain:** Identify any restrictions on x and y
2. **Rewrite the inequality:** Move all terms to one side, to get $f(x, y) \leq 0$.
3. **Find zeroes:** Solve $f(x, y) = 0$ to determine the dividing line or curve.
4. **Identify the solution region:** Pick a test point in each region and evaluate $f(x, y)$ to determine where the inequality holds.

✓ Solution 3: (a) Solving the inequality $2x + 6 \leq 3y$ **Step 1: Find the domain**

The given function is a linear equation, which is continuous and defined for all $(x, y) \in \mathbb{R}^2$. Hence, the domain is:

$$\mathbb{R}^2$$

Step 2: Rewrite the inequality

We express the inequality in the form $f(x, y) \leq 0$:

$$2x + 6 - 3y \leq 0$$

This means we analyze the sign of the function:

$$f(x, y) = 2x + 6 - 3y$$

Step 3: Find zeroes

Setting $f(x, y) = 0$:

$$2x + 6 - 3y = 0$$

Solving for y :

$$y = \frac{2}{3}x + 2$$

This is a straight line that separates the plane into two regions. To determine which side of the line satisfies the inequality, pick a test point and check its sign.

Step 4: Identify the solution region

We select a test point to determine the sign of $f(x, y)$ in different regions.

- Test Point: $(0, 0)$

$$f(0, 0) = 2(0) + 6 - 3(0) = 6 > 0$$

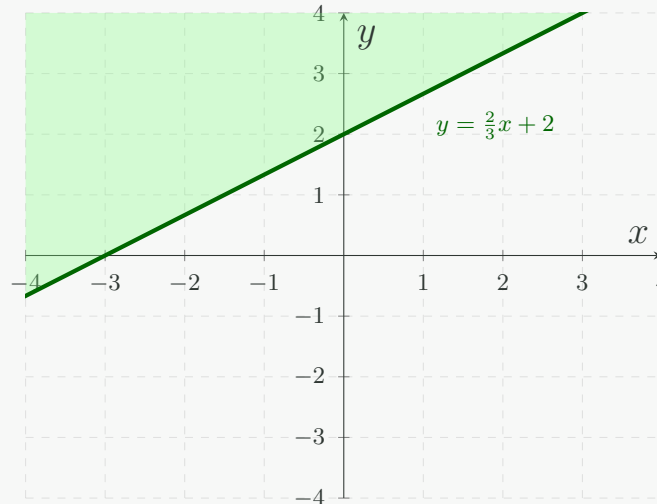
Since $f(0, 0) > 0$, the region containing $(0, 0)$ is not part of the solution.

- Test Point: $(-3, 3)$

$$f(-3, 3) = 2(-3) + 6 - 3(3) = -6 + 6 - 9 = -9 < 0$$

Since $f(-3, 3) < 0$, this region satisfies the inequality.

Answer: The green region represents where the inequality $f(x, y) \leq 0$ holds, and the dark green line represents the boundary where $f(x, y) = 0$.



✓ Solution 3: (b) Solving the inequality $y > y \log_2 x$

Step 1: Find the domain

Since the function involves $\log_2 x$, we must ensure that $x > 0$ because the logarithm is only defined for positive values of x . Thus, the domain is:

$$\{(x, y) \in \mathbb{R}^2 \mid x > 0\}$$

Step 2: Rewrite the inequality

Rewriting the given inequality:

$$y > y \log_2 x$$

Factor out y :

$$y(1 - \log_2 x) > 0$$

This inequality is satisfied when:

- $y > 0$ and $1 - \log_2 x > 0$ (i.e., $x < 2$)
- $y < 0$ and $1 - \log_2 x < 0$ (i.e., $x > 2$)

Step 3: Find zeroes

Setting $f(x, y) = 0$:

$$y \cdot (1 - \log_2 x) = 0$$

This occurs when either:

$$y = 0 \quad \text{or} \quad x = 2$$

Thus, the boundary consists of the x-axis and the vertical line $x = 2$.

Step 4: Identify the solution region

We select test points to determine the sign of $f(x, y)$ in different regions.

- Test Point 1: $(1, 2)$ (left of $x = 2$, above $y = 0$)

$$f(1, 2) = 2(1 - \log_2 1) = 2(1 - 0) = 2 > 0$$

Hence, this region is positive (green area).

- Test Point 2: $(3, 2)$ (right of $x = 2$, above $y = 0$)

$$f(3, 2) = 2(1 - \log_2 3) = 2(1 - \log_2 3) < 0$$

Since $\log_2 3 > 1$, this region is negative (red boundary).

- Test Point 3: $(1, -2)$ (left of $x = 2$, below $y = 0$)

$$f(1, -2) = (-2)(1 - \log_2 1) = -2(1 - 0) = -2 < 0$$

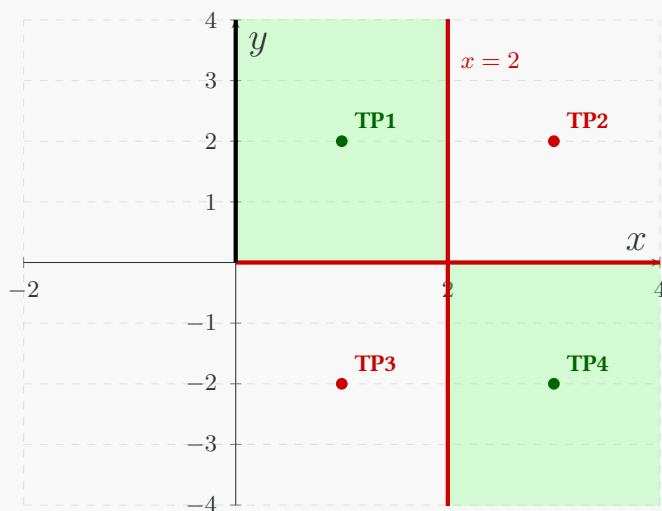
Hence, this region is negative (red boundary).

- Test Point 4: $(3, -2)$ (right of $x = 2$, below $y = 0$)

$$f(3, -2) = (-2)(1 - \log_2 3) = -2(1 - \log_2 3) > 0$$

Since $\log_2 3 > 1$, this region is positive (green area).

Answer: The green region represents where the function is positive, red lines indicate where the function is zero, and black lines indicate where the function is undefined .



? Question 4

Graph the contour lines of levels -1 , -0.5 , 0 , 0.5 , and 1 of the function

$$f(x, y) = \frac{1}{x + y - 2}$$

on the (x, y) -plane.

💡 Tip: Steps for Finding Contour Lines

To find contour lines of a function $f(x, y)$, follow these steps:

1. **Find the domain** – Determine where $f(x, y)$ is defined.
2. **Set up the contour equation** – Solve $f(x, y) = a$ for a general constant a .
3. **Check when it exists** – Identify which values of a lead to valid contour lines.
4. **Draw the contour lines** – Plot the contour lines for the given levels.

✓ Solution 4: Finding the contour lines of $f(x, y) = \frac{1}{x+y-2}$ **Step 1: Find the domain**

The function $f(x, y) = \frac{1}{x+y-2}$ is undefined where the denominator is zero:

$$x + y - 2 = 0 \quad \Rightarrow \quad y = 2 - x$$

Thus, the domain is:

$$\{(x, y) \in \mathbb{R}^2 \mid x + y \neq 2\}$$

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$ for given values of a :

$$\frac{1}{x+y-2} = a \Rightarrow a(x+y-2) = 1$$

Step 3: Check for existence

If $a = 0$, the equation is not defined, so no contour line exists. If $a \neq 0$, we can divide by a and solve for y :

$$y = -x + 2 + \frac{1}{a}$$

Step 4: Plot the contour lines

We now plot the contour lines for different values of a :

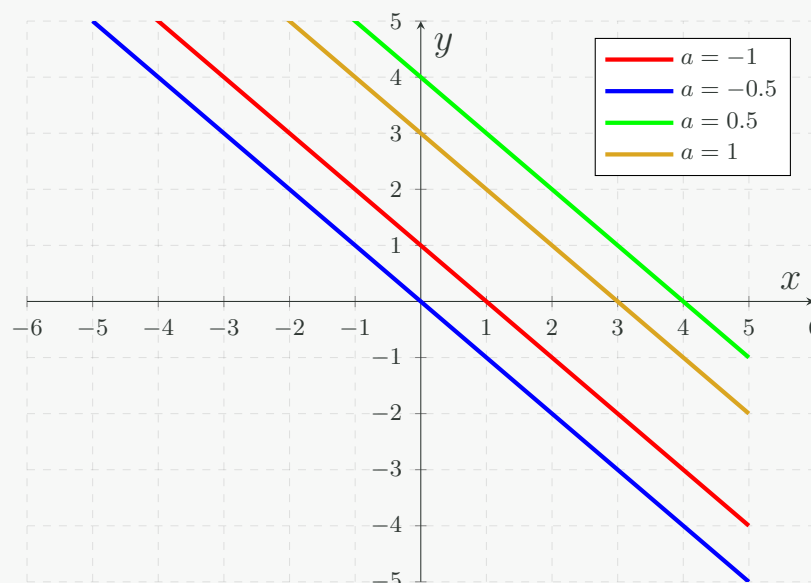
$$a = -1 \Rightarrow y = -x + 2 - 1 \Rightarrow y = -x + 1$$

$$a = -0.5 \Rightarrow y = -x + 2 - 2 \Rightarrow y = -x$$

$$a = 0.5 \Rightarrow y = -x + 2 + 2 \Rightarrow y = -x + 4$$

$$a = 1 \Rightarrow y = -x + 2 + 1 \Rightarrow y = -x + 3$$

Since $a = 0$ leads to an undefined equation, no contour line exists for this case.



Answer: The contour lines are given by $y = -x + 2 + \frac{1}{a}$, where $a \neq 0$. There is no contour line for $a = 0$ since the function is undefined in that case.

? Question 5

Graph the contour lines of levels $-2, -1, 0, 1, 2$ of the function

$$f(x, y) = \sqrt{y - x + 1}$$

on the (x, y) -plane.

✓ Solution 5: Finding the contour lines of $f(x, y) = \sqrt{y - x + 1}$ **Step 1: Find the domain**

The function $f(x, y) = \sqrt{y - x + 1}$ involves a square root, which requires the argument to be non-negative:

$$y - x + 1 \geq 0 \quad \Rightarrow \quad y \geq x - 1$$

Thus, the domain is:

$$\{(x, y) \in \mathbb{R}^2 \mid y \geq x - 1\}$$

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$ for given values of a :

$$\sqrt{y - x + 1} = a \quad \Rightarrow \quad a^2 = y - x + 1$$

Step 3: Check for existence

If $a < 0$, the equation does not exist since the square root function is non-negative. If $a \geq 0$, we can solve for y :

$$y = x - 1 + a^2$$

Step 4: Plot the contour lines

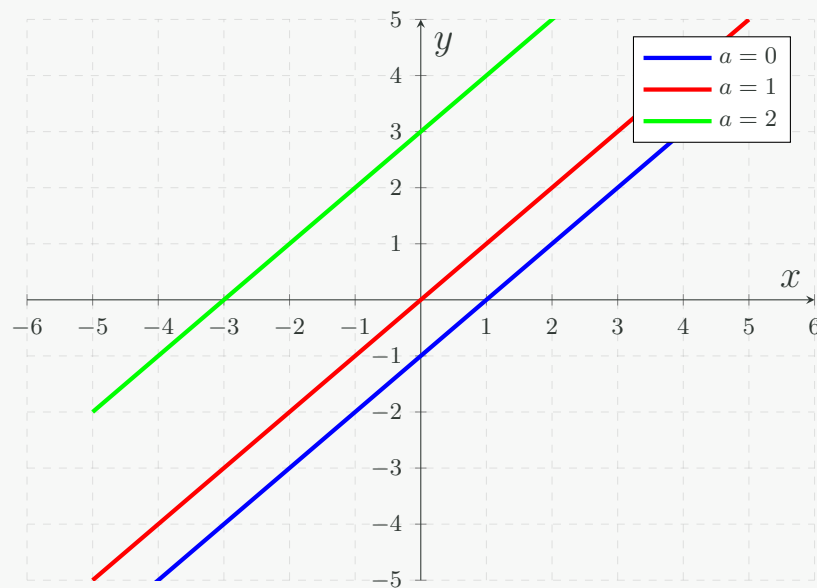
We now plot the contour lines for different values of a :

$$a = 0 \quad \Rightarrow \quad y = x - 1$$

$$a = 1 \quad \Rightarrow \quad y = x - 1 + 1 \quad \Rightarrow \quad y = x$$

$$a = 2 \quad \Rightarrow \quad y = x - 1 + 4 \quad \Rightarrow \quad y = x + 3$$

Since $a < 0$ is not possible, no contour lines exist for those cases.



Answer: The contour lines are given by $y = x - 1 + a^2$, where $a \geq 0$. There are no contour lines for $a < 0$ since the function is not defined for those values.

? Question 6

Graph the contour line of level 4 of the function

$$f(x, y) = \log_2(x^2 + y^2 - 9)$$

on the (x, y) -plane.

✓ Solution 6: Finding the contour line of level 4 of $f(x, y) = \log_2(x^2 + y^2 - 9)$ **Step 1: Find the domain**

The function $f(x, y) = \log_2(x^2 + y^2 - 9)$ is defined when its argument is positive:

$$x^2 + y^2 - 9 > 0 \quad \Rightarrow \quad x^2 + y^2 > 9$$

Thus, the domain consists of all points outside the circle $x^2 + y^2 = 9$.

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$ for given values of a :

$$\log_2(x^2 + y^2 - 9) = a$$

Rearranging:

$$x^2 + y^2 - 9 = 2^a$$

Step 3: Check for existence

For the equation to be valid, the right-hand side must be positive, meaning:

$$2^a > 0$$

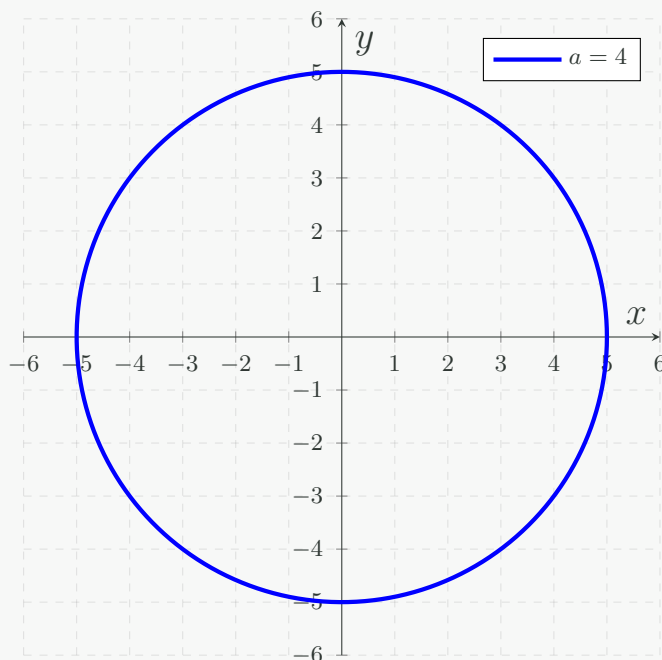
Since 2^a is always positive for all real a , contour lines exist for all values of a .

Step 4: Plot the contour lines

We now plot the contour line for $a = 4$:

$$2^4 = x^2 + y^2 - 9 \Rightarrow x^2 + y^2 = 25$$

which represents a circle centered at $(0, 0)$ with radius 5.



Answer: The contour line for $a = 4$ is given by $x^2 + y^2 = 25$, which is a circle of radius 5 centered at $(0, 0)$. Contour lines exist for all real a .

? Question 7

Graph the contour lines of levels $-4, -2, 0, 2, 4$ of the following functions:

(a) $f(x, y) = x + y$

(b) $f(x, y) = x - y$

(c) $f(x, y) = x$

(d) $f(x, y) = x^2 + y^2$

(e) $f(x, y) = (x - 2)^2 + y^2$

(f) $f(x, y) = x^2 + (y + 2)^2$

✓ Solution 7: (a) Finding the contour lines of $f(x, y) = x + y$ **Step 1: Find the domain**

The function $f(x, y) = x + y$ is defined everywhere in $\mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$ for given values of a :

$$x + y = a \quad \Rightarrow \quad y = -x + a$$

Step 3: Check for existence

Since this equation is valid for all real numbers a , contour lines exist for every level.

Step 4: Plot the contour lines

We now plot the contour lines for different values of a :

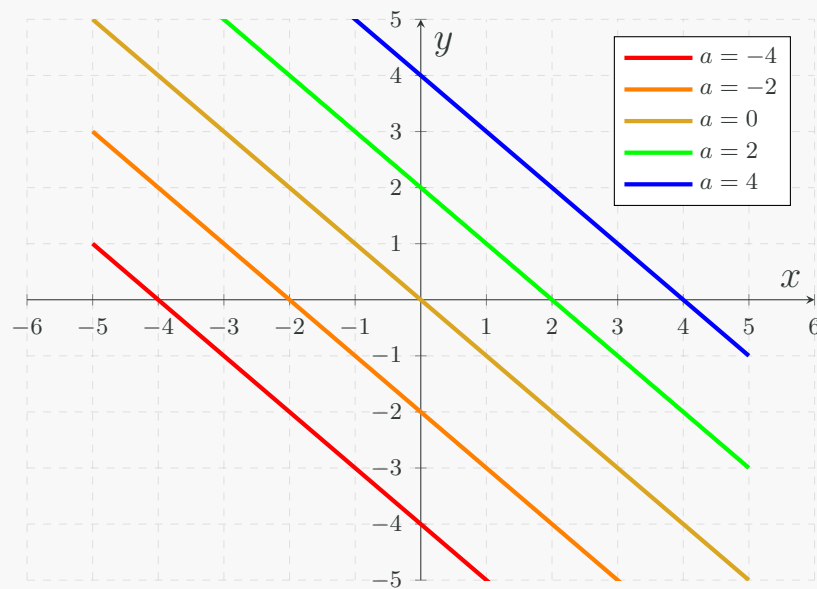
$$a = -4 \Rightarrow y = -x - 4$$

$$a = -2 \Rightarrow y = -x - 2$$

$$a = 0 \Rightarrow y = -x$$

$$a = 2 \Rightarrow y = -x + 2$$

$$a = 4 \Rightarrow y = -x + 4$$



Answer: The contour lines are given by $y = -x + a$ for all real values of a .

✓ Solution 7: (b) Finding the contour lines of $f(x, y) = x - y$

Step 1: Find the domain

The function $f(x, y) = x - y$ is a linear function and is defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$, so:

$$x - y = a \quad \Rightarrow \quad y = x - a$$

Step 3: Check for existence

Since this is a linear function defined everywhere, the contour lines exist for all $a \in \mathbb{R}$.

Step 4: Plot the contour lines

We now plot the contour lines for different values of a :

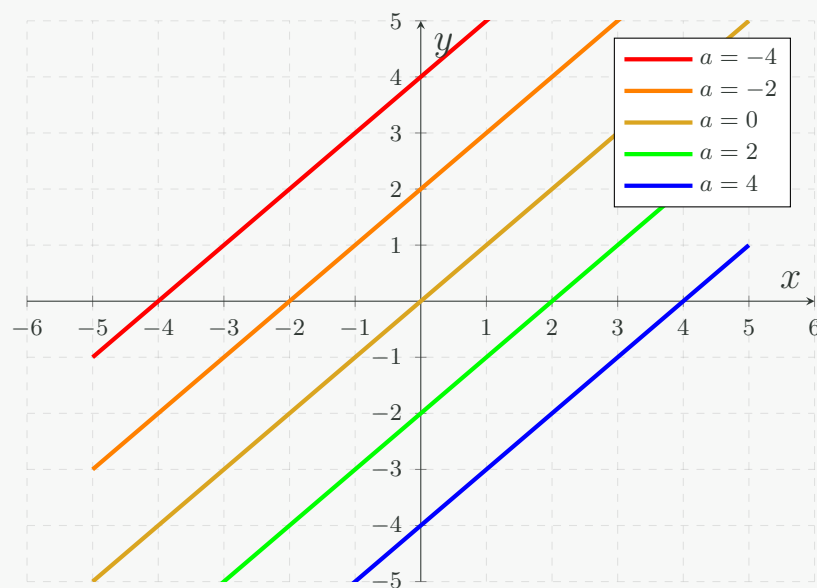
$$a = -4 \quad \Rightarrow \quad y = x + 4$$

$$a = -2 \quad \Rightarrow \quad y = x + 2$$

$$a = 0 \quad \Rightarrow \quad y = x$$

$$a = 2 \quad \Rightarrow \quad y = x - 2$$

$$a = 4 \quad \Rightarrow \quad y = x - 4$$



Answer: The contour lines are given by $y = x - a$ and exist for all $a \in \mathbb{R}$.

✓ **Solution 7: (c) Finding the contour lines of $f(x, y) = x$**

Step 1: Find the domain

The function $f(x, y) = x$ is a linear function and is defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$, so:

$$x = a$$

Step 3: Check for existence

Since this is a linear function defined everywhere, the contour lines exist for all $a \in \mathbb{R}$.

Step 4: Plot the contour lines

We now plot the contour lines for different values of a :

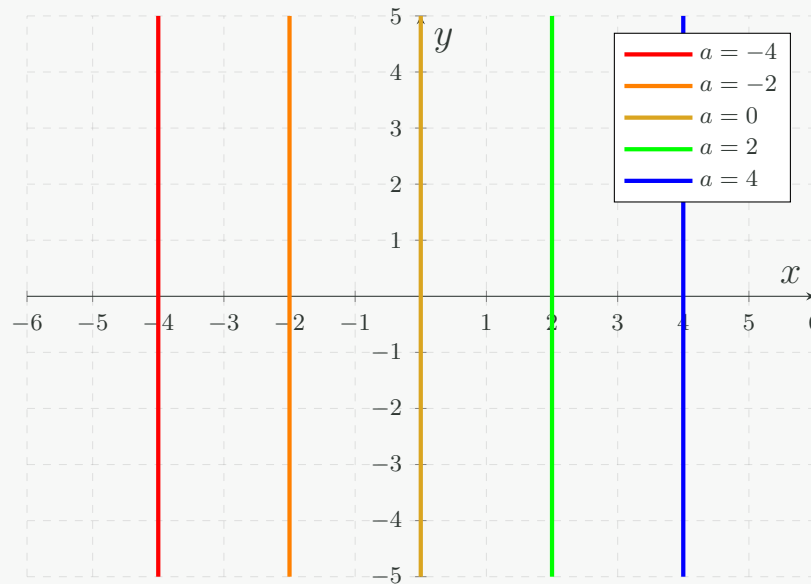
$$a = -4 \quad \Rightarrow \quad x = -4$$

$$a = -2 \quad \Rightarrow \quad x = -2$$

$$a = 0 \quad \Rightarrow \quad x = 0$$

$$a = 2 \quad \Rightarrow \quad x = 2$$

$$a = 4 \quad \Rightarrow \quad x = 4$$



Answer: The contour lines are given by $x = a$ and exist for all $a \in \mathbb{R}$.

✓ Solution 7: (d) Finding the contour lines of $f(x, y) = x^2 + y^2$

Step 1: Find the domain

The function $f(x, y) = x^2 + y^2$ is a polynomial and is defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$, so:

$$x^2 + y^2 = a$$

Step 3: Check for existence

Since $x^2 + y^2 \geq 0$ for all (x, y) , contour lines only exist for $a \geq 0$. If $a < 0$, no real solutions exist, meaning no contour lines are drawn for such values.

Step 4: Plot the contour lines

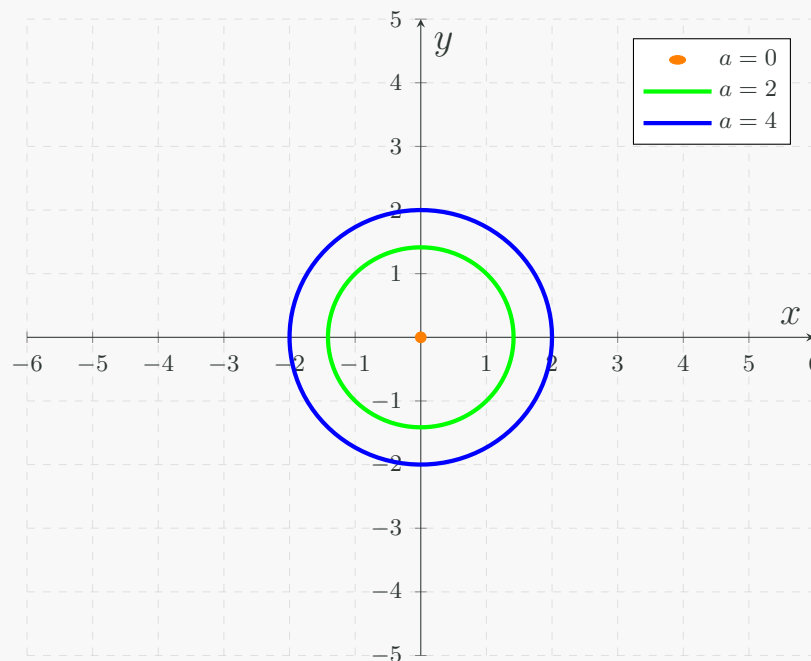
We now plot the contour lines for different values of a :

$$a = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow (0, 0) \text{ (single point)}$$

$$a = 2 \Rightarrow x^2 + y^2 = 2$$

$$a = 4 \Rightarrow x^2 + y^2 = 4$$

Since $a = -4$ and $a = -2$ result in no valid real solutions, no contour lines exist for those levels.



Answer: The contour lines are given by $x^2 + y^2 = a$ and exist for $a \geq 0$. There are no contour lines for $a < 0$ since the equation has no real solutions.

✓ Solution 7: (e) Finding the contour lines of $f(x, y) = (x - 2)^2 + y^2$ **Step 1: Find the domain**

The function is a quadratic polynomial in x and y , which means it is defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$ for given values of a :

$$(x - 2)^2 + y^2 = a$$

Step 3: Check for existence

Since the left-hand side represents the sum of squares, the equation is only valid when $a \geq 0$. There are no contour lines for negative values of a .

Step 4: Plot the contour lines

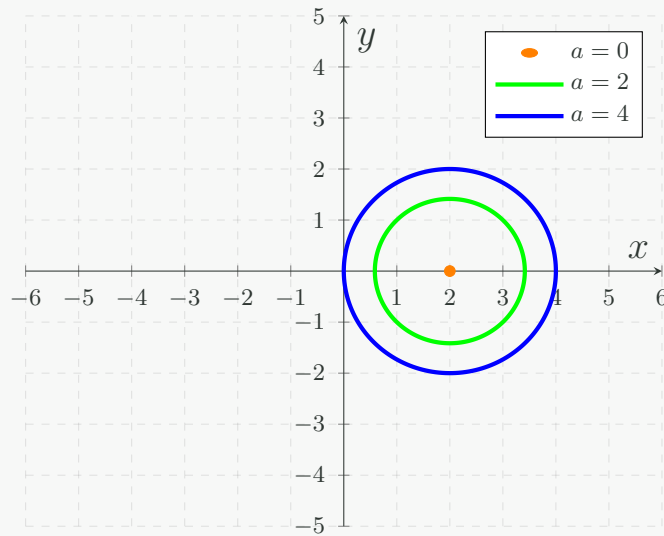
We now plot the contour lines for different values of a :

$$a = 0 \Rightarrow (x - 2)^2 + y^2 = 0 \Rightarrow (2, 0) \text{ (single point)}$$

$$a = 2 \Rightarrow (x - 2)^2 + y^2 = 2 \Rightarrow \text{Circle of radius } \sqrt{2} \text{ centered at } (2, 0)$$

$$a = 4 \Rightarrow (x - 2)^2 + y^2 = 4 \Rightarrow \text{Circle of radius } 2 \text{ centered at } (2, 0)$$

There are no contour lines for levels -4 and -2 , since there are only points satisfying the equation when $a \geq 0$.



Answer: The contour lines are given by $(x-2)^2 + y^2 = a$, where $a \geq 0$. There are no contour lines for $a < 0$, since the sum of squares is always non-negative.

✓ Solution 7: (f) Finding the contour lines of $f(x, y) = x^2 + (y + 2)^2$

Step 1: Find the domain

The function is a quadratic polynomial in x and y , which means it is defined for all $(x, y) \in \mathbb{R}^2$, so the domain is:

$$\mathbb{R}^2$$

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$ for given values of a :

$$x^2 + (y + 2)^2 = a$$

Step 3: Check for existence

Since the left-hand side represents the sum of squares, the equation is only valid when $a \geq 0$. There are no contour lines for negative values of a .

Step 4: Plot the contour lines

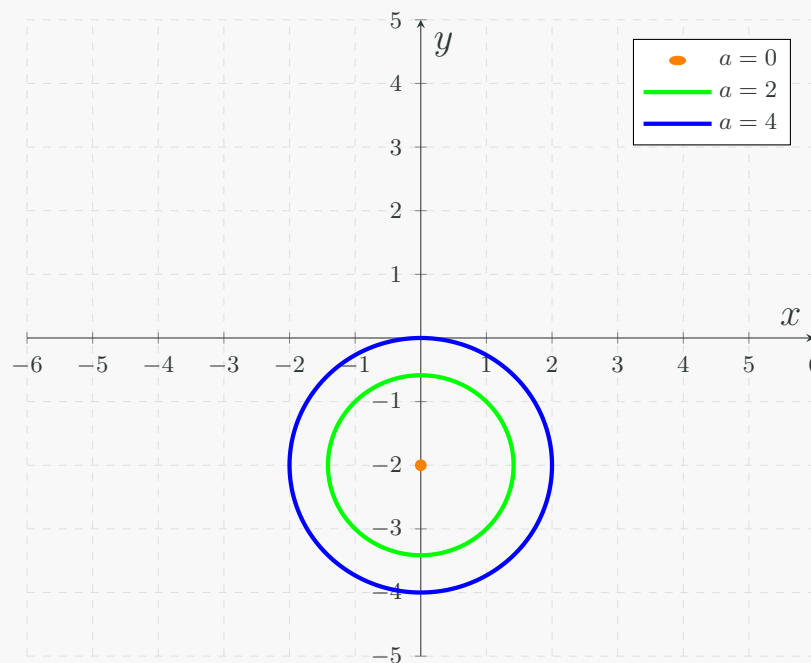
We now plot the contour lines for different values of a :

$$a = 0 \Rightarrow x^2 + (y + 2)^2 = 0 \Rightarrow (0, -2) \text{ (single point)}$$

$$a = 2 \Rightarrow x^2 + (y + 2)^2 = 2 \Rightarrow \text{Circle of radius } \sqrt{2} \text{ centered at } (0, -2)$$

$$a = 4 \Rightarrow x^2 + (y + 2)^2 = 4 \Rightarrow \text{Circle of radius } 2 \text{ centered at } (0, -2)$$

There are no contour lines for levels -4 and -2 , since there are only points satisfying the equation when $a \geq 0$.



Answer: The contour lines are given by $(x - 2)^2 + y^2 = a$, where $a \geq 0$. There are no contour lines for $a < 0$, since the sum of squares is always non-negative.

? Question 8

The figure below contains graphs of the functions listed in question 7. Use the contour lines you found in question 7 to determine which graph corresponds to which function.

Note: Refer to the original exercise document for graphs.

💡 Tip: Matching 3D Graphs with Contour Lines

When looking at contour lines, think of them as "slices" of the 3D surface at different heights. If you take a 3D graph and slice it with a horizontal plane at $z = a$, the shape of the intersection is the contour line for that value.

- If the 3D graph is a **plane**, the intersection will always be a **straight line**.
- If the 3D graph is a **curved surface** (like a paraboloid or sphere), the intersection will be a **curve**, usually a circle.

✓ Solution 8: Matching 3D graphs with contour plots from question 7

First, observe that the 3D graphs $8a$, $8b$, and $8e$ represent planes. Since slicing a plane with a horizontal plane $z = a$ always results in straight lines, their corresponding contour plots must also contain only straight lines. This means we are looking to match them with contour plots from $7a$, $7b$, and $7c$, which consist of straight lines.

On the other hand, the 3D graphs $8c$, $8d$, and $8f$ represent curved surfaces. Slicing these surfaces produces circles. Therefore, we need to match them with contour plots from $7d$, $7e$, and $7f$, which consist of concentric circles.

Step 1: Match the 3D planes with straight-line contour plots

To determine which plane corresponds to which contour plot, we analyze the direction of the lines. The easiest way to do this is by slicing the 3D plane at $z = 0$ and checking the resulting line.

- Graph 8a has lines that are parallel to the y -axis and orthogonal to the x -axis, which matches the contour lines in 7c.
- Graph 8b has diagonal lines that are increasing from left to right. Be careful when analyzing the image, as the 3D perspective can sometimes be misleading, but observing the slope confirms that it matches 7b.
- By elimination, 8e must correspond to 7a. To confirm, we observe that its diagonal lines are decreasing from left to right, which aligns with the contour plot in 7a.

Step 2: Match the curved 3D surfaces with circular contour plots

For the curved graphs, the key is to check where the center of the circles is located.

- Graph 8c has circles that are centered to the right, which matches the contour lines in 7e.
- Graph 8d has circles that are centered at $(0, 0)$, so it must be 7d.
- Graph 8f looks similar to 8c but with the center shifted downward, so it must be 7f.

Answer:

8a	8b	8c	8d	8e	8f
7c	7b	7e	7d	7a	7f

? Question 9

The images below are the graphs of three functions of two variables (we see only a part of the graph, the graphs continue indefinitely.)

Sketch a contour plot for each of the three functions.

Note: Refer to the original exercise document for graphs.

💡 Tip: Understanding Parabolic Contours

When a 3D surface represents a parabolic shape (saddle), slicing it with horizontal planes at different heights produces contour lines in the shape of parallel curves. The closer the curves are to each other, the steeper the surface in that region.

✅ Solution 9: (a) Contour plot for 3D graph (a)

Here contour lines appear as curved parallel lines. These curves open outward because the surface increases in both positive and negative directions. The spacing between curves shows how quickly the function value changes.

Note: Refer to the original solution document for contour map.

💡 Tip: Understanding Circular Contours

When a 3D surface is radially symmetric (such as a peak or a bowl), slicing it at different heights results in concentric circular contour lines. The closer the circles are, the steeper the function at that region.

✓ Solution 9: (b) Contour plot for 3D graph (b)

The 3D graph in (b) represents a radially symmetric peak, meaning its contour plot consists of concentric circles. As the height increases, the circles get smaller, showing that the function values are highest at the center and decrease outward.

Note: Refer to the original solution document for contour map.

💡 Tip: Understanding Linear Contours

When a 3D surface represents a plane, slicing it with horizontal planes produces straight contour lines. The direction of these lines corresponds to the orientation of the 3D plane.

✓ Solution 9: (c) Contour plot for 3D graph (c)

The 3D graph in (c) represents a tilted plane, meaning that slicing it at different heights produces straight, parallel lines. These lines indicate that the surface changes linearly with no curvature.

Note: Refer to the original solution document for contour map.

? Question 10

Consider the function f given by the equation

$$f(x, y) = -x^2 + y.$$

- (a) Find the equation of a general contour line of the function f (that is, give the equation of the contour line of level a , where a stands for any real number).
- (b) Give an equation of the contour lines of levels $-1, 0, 1$ and graph these contour lines.
- (c) For which values of a is the contour line non-empty?
- (d) Find the range of the function f .

💡 Tip: Steps for Finding Contour Lines

To find contour lines of a function $f(x, y)$, follow these steps:

1. **Find the domain** – Determine where $f(x, y)$ is defined.
2. **Set up the contour equation** – Solve $f(x, y) = a$ for a general constant a .
3. **Check when it exists** – Identify which values of a lead to valid contour lines.
4. **Draw the contour lines** – Plot the contour lines for the given levels.

✓ Solution 10: (a) General equation of contour lines of $f(x, y) = -x^2 + y$

The function $f(x, y) = -x^2 + y$ is a polynomial and is therefore defined for all $(x, y) \in \mathbb{R}^2$.

Contour lines correspond to solving $f(x, y) = a$ for a given level a :

$$-x^2 + y = a \quad \Rightarrow \quad y = x^2 + a$$

Answer: $y = x^2 + a$

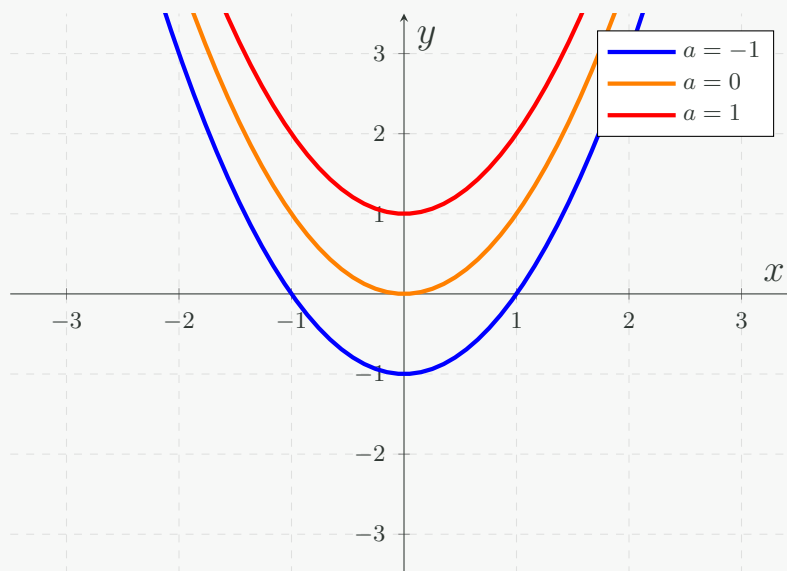
✓ **Solution 10: (b) Contour lines of levels $-1, 0, 1$ of $f(x, y) = -x^2 + y$**

Answer: Setting a to the given values:

$$a = -1 \Rightarrow y = x^2 - 1,$$

$$a = 0 \Rightarrow y = x^2,$$

$$a = 1 \Rightarrow y = x^2 + 1.$$



✓ **Solution 10: (c) Values of a for which contour lines of $f(x, y) = -x^2 + y$ are non-empty**

Since the equation $y = x^2 + a$ is valid for all real numbers, contour lines exist for all real values of a .

Answer: $\mathbb{R}$

✓ Solution 10: (d) Range of $f(x, y) = -x^2 + y$

The function is given by:

$$f(x, y) = -x^2 + y$$

For any fixed x , the function behaves as a linear equation in y , meaning it can take all real values. Since there are no restrictions on y , the range of f is

$$\mathbb{R}$$

.

This directly connects to part (c), where we found that contour lines exist for all values of a . Since contour lines represent level curves of $f(x, y)$, the fact that every a produces a valid contour line confirms that $f(x, y)$ can attain every real number as an output.

Answer: $\mathbb{R}$

? Question 11

The function f is given by an equation of the form

$$f(x, y) = ax^2 + by,$$

where a and b are real numbers. The contour line of level 5 contains the point $(1, 1)$ and the contour line of level 8 contains the point $(1, 2)$.

Find the numbers a and b .

✓ Solution 11: Finding the coefficients a and b of $f(x, y) = ax^2 + by$ **Step 1: Set up the system of equations**

Since the function is given as

$$f(x, y) = ax^2 + by,$$

we substitute the given points into the equation.

For the contour line of level 5 containing $(1, 1)$:

$$5 = a(1)^2 + b(1) \quad \Rightarrow \quad 5 = a + b.$$

For the contour line of level 8 containing $(1, 2)$:

$$8 = a(1)^2 + b(2) \quad \Rightarrow \quad 8 = a + 2b.$$

Step 2: Solve for a and b

We now solve the system of equations:

$$a + b = 5,$$

$$a + 2b = 8.$$

Express a from the second equation:

$$a = 8 - 2b.$$

Substituting into the first equation:

$$(8 - 2b) + b = 5 \Rightarrow 8 - b = 5 \Rightarrow b = 3.$$

Substituting $b = 3$ into $a = 8 - 2b$:

$$a = 8 - 2(3) \Rightarrow a = 2.$$

Answer: $a = 2$, $b = 3$.

? Question 12

Consider the function given by

$$f(x, y) = \frac{\ln(1 - x^2)}{\ln(y)}.$$

- (a) Find the domain of f .
- (b) Graph the contour lines of levels 0, 1 and 2 for this function.
- (c) Find the range of f .

💡 Tip: Understanding Domain Restrictions

To determine the domain of a function, check for the following:

- **Square roots:** The expression inside must be non-negative.
- **Logarithms:** The argument must be positive.
- **Denominators:** The denominator must not be zero.

Applying these rules ensures the function is well-defined.

✓ Solution 12: (a) Find the domain of $f(x, y) = \frac{\ln(1 - x^2)}{\ln(y)}$

The function is given by:

$$f(x, y) = \frac{\ln(1 - x^2)}{\ln(y)}$$

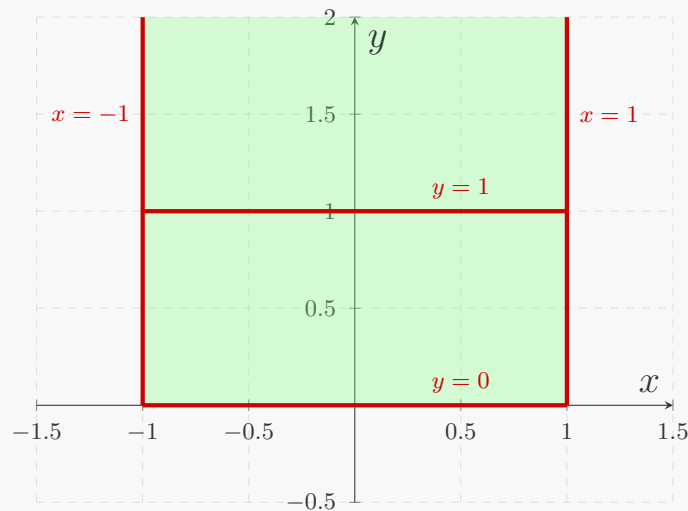
Applying the domain restrictions:

- The logarithm in the numerator requires $1 - x^2 > 0$, which simplifies to $-1 < x < 1$.
- The logarithm in the denominator requires $y > 0$.

- The denominator must not be zero, so $\ln(y) \neq 0$, which means $y \neq 1$.

Thus, the domain of $f(x, y)$ is:

$$\{(x, y) \in \mathbb{R}^2 \mid -1 < x < 1, \quad y > 0, \quad y \neq 1\}$$



Answer: The domain is $-1 < x < 1$, $y > 0$, and $y \neq 1$. The valid region is shaded in green, and excluded boundaries are marked in red.

Tip: Steps for Graphing Contour Lines

To graph contour lines of a function $f(x, y)$, follow these steps:

1. **Find the domain** – Identify where $f(x, y)$ is defined. This was determined in part (a).
2. **Set up the contour equation** – Solve $f(x, y) = a$ for a general constant a .
3. **Check when it exists** – Determine for which values of a the equation produces valid contour lines.
4. **Draw the contour lines** – Plot the contour lines for the given levels.

✓ Solution 12: (b) Contour lines of levels 0, 1, 2 of $f(x, y) = \frac{\ln(1-x^2)}{\ln(y)}$

Step 1: Find the domain

From part (a), the domain is $-1 < x < 1$, $y > 0$, and $y \neq 1$.

Step 2: Set up the contour equation

Contour lines correspond to solving $f(x, y) = a$ for given values of a :

$$\frac{\ln(1-x^2)}{\ln(y)} = a \quad \Rightarrow \quad \ln(1-x^2) = a \ln(y)$$

Now, exponentiate both sides:

$$e^{\ln(1-x^2)} = e^{a \ln(y)} \quad \Rightarrow \quad 1-x^2 = y^a$$

At this point, solving for y requires checking whether $a \neq 0$:

- If $a \neq 0$, we take the $\frac{1}{a}$ -th power on both sides and solve:

$$y = (1-x^2)^{\frac{1}{a}}$$

- If $a = 0$, the equation simplifies to $\ln(1-x^2) = 0$, which means:

$$1-x^2 = 1 \quad \Rightarrow \quad x^2 = 0 \quad \Rightarrow \quad x = 0$$

This represents the y -axis.

Step 3: Check for existence

Now, we analyze specific contour levels a :

$$a = 0 \quad \Rightarrow \quad x = 0, \quad \text{which is the } y\text{-axis.}$$

$$a = 1 \quad \Rightarrow \quad y = 1-x^2, \quad \text{a parabola with vertex at } (0, 1)$$

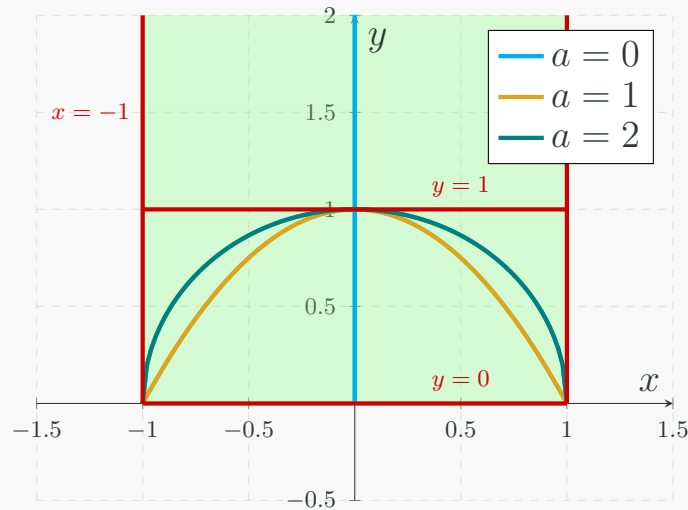
and passing through $(-1, 0)$ and $(1, 0)$

$$a = 2 \quad \Rightarrow \quad x^2 + y^2 = 1, \quad \text{a circle centered at } (0, 0) \text{ with radius } 1.$$

Keep in mind that all calculations must also respect the domain constraints.

Step 4: Draw the contour lines

Answer:



💡 Tip: Understanding the Range of a Function

The range of a function represents all possible output values. To determine this, we analyze the function's behavior over its entire domain and check what values $z = f(x, y)$ can attain.

One way to determine the range is by examining contour lines. Each contour line corresponds to a level curve where $f(x, y) = a$. If contour lines exist for all real values of a , then the function can take any real number as an output, meaning its range is $\mathbb{R}$. Conversely, if some values of a do not produce valid contour lines, those values are not in the range.

✓ **Solution 12: (c)** Find the range of $f(x, y) = \frac{\ln(1-x^2)}{\ln(y)}$

In part (b), we saw that no matter what value of a we choose, there is always a valid contour line within the domain. This means that for any real number a , we can always find some pair (x, y) where $f(x, y) = a$. Since this works for every a , the function can take all real values.

Answer: The range of $f(x, y)$ is $\mathbb{R}$.

? Question 13

A company produces two products. The amount produced of product 1 and product 2 is given by q_1 and q_2 respectively.

- The production cost for one unit of product 1 is 6 monetary units.
- The production cost for one unit of product 2 is 5 monetary units.
- The fixed cost is 269 monetary units.
- The inverse demand function for the first product is given by $p_1 = 30 - q_1$.
- The inverse demand function for the second product is given by $p_2 = 35 - q_2$.

Make a figure which shows which combinations of q_1 and q_2 yield profit.

💡 Tip: Key economic terms

- 1. Variables:** q_1 and q_2 represent the quantities produced, p_1 and p_2 are the selling prices, and c_1 and c_2 are the production costs per unit.
- 2. Inverse Demand Function:** $p_1 = f(q_1)$, $p_2 = g(q_2)$.
- 3. Revenue Function:** $R(q_1, q_2) = p_1 \cdot q_1 + p_2 \cdot q_2$.
- 4. Cost Function:** $C(q_1, q_2) = c_1 \cdot q_1 + c_2 \cdot q_2 + C_{fixed}$.
- 5. Profit Function:** $P(q_1, q_2) = R(q_1, q_2) - C(q_1, q_2)$.

Tip: Solving Profit Inequalities

When determining which combinations of q_1 and q_2 yield profit, we aim to find where $P(q_1, q_2) > 0$. So that just solving inequality we can follow steps that were named before when solving inequalities (such as question 3)

To remind and to be more specific, here's what we can do. Follow these structured steps:

Step 1: Express the Profit Function: Formulate $P(q_1, q_2)$ in terms of q_1 and q_2

Step 2: Determine the Domain

1. **Mathematical Constraints:** Ensure the function is well-defined

2. **Application Constraints:** In real-world scenarios:

- Quantities: $q_1, q_2 \geq 0$,
- Prices: $p_1, p_2 \geq 0$

Step 3: Solve the Inequality: $P(q_1, q_2) > 0$

Step 4: Visualize the Solution: Shade the areas where $P(q_1, q_2)$ is positive.

Solution 13: Finding which combinations of q_1 and q_2 yield profit

Step 0: Identify Given Information

Before solving the problem, we list the key values and equations provided in the question:

- Production Costs: $c_1 = 6$, $c_2 = 5$, $C_{\text{fixed}} = 269$.
- Inverse Demand Functions: $p_1 = 30 - q_1$, $p_2 = 35 - q_2$.

Step 1: Express the Profit Function

The total revenue is found by multiplying quantity by price for each product:

$$R(q_1, q_2) = q_1 p_1 + q_2 p_2 = q_1(30 - q_1) + q_2(35 - q_2).$$

Expanding:

$$R(q_1, q_2) = 30q_1 - q_1^2 + 35q_2 - q_2^2.$$

The total cost includes variable and fixed costs:

$$C(q_1, q_2) = 6q_1 + 5q_2 + 269.$$

Thus, the profit function is:

$$P(q_1, q_2) = R(q_1, q_2) - C(q_1, q_2).$$

Substituting:

$$P(q_1, q_2) = (30q_1 - q_1^2 + 35q_2 - q_2^2) - (6q_1 + 5q_2 + 269).$$

Simplifying:

$$P(q_1, q_2) = -q_1^2 - q_2^2 + 24q_1 + 30q_2 - 269.$$

Step 2: Determine the Domain

To find valid values for q_1 and q_2 , we analyze both mathematical and application-based constraints.

- **Mathematical Constraints:** The profit function $P(q_1, q_2)$ is a quadratic expression in q_1 and q_2 , which is well-defined for all real numbers.
- **Application Constraints:**

$$q_1 \geq 0, \quad q_2 \geq 0.$$

$$p_1 = 30 - q_1 \geq 0 \quad \Rightarrow \quad q_1 \leq 30,$$

$$p_2 = 35 - q_2 \geq 0 \quad \Rightarrow \quad q_2 \leq 35.$$

Thus, the domain of $P(q_1, q_2)$ is:

$$0 \leq q_1 \leq 30, \quad 0 \leq q_2 \leq 35.$$

Step 3: Solving the Inequality

We need to solve:

$$P(q_1, q_2) = -q_1^2 - q_2^2 + 24q_1 + 30q_2 - 269 > 0.$$

💡 Tip: Completing the Square

Completing the square helps rewrite quadratic expressions in a structured form, revealing standard shapes like circles.

Completing the square for both q_1 and q_2 :

$$-(q_1^2 - 24q_1) - (q_2^2 - 30q_2) > 269.$$

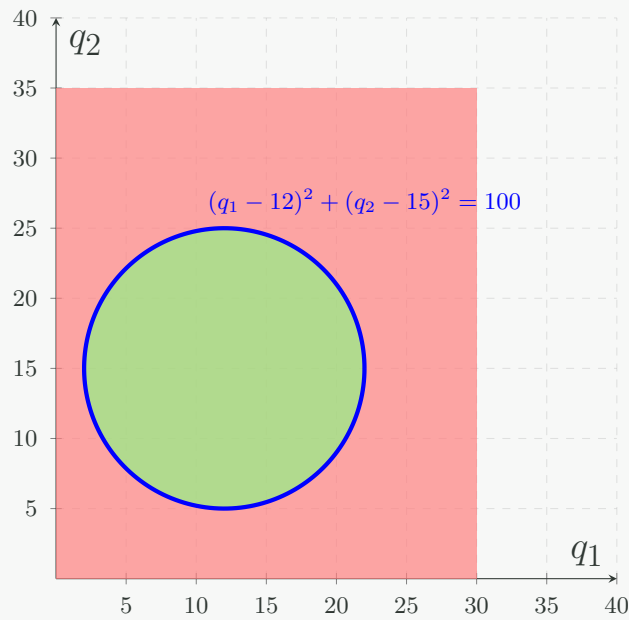
Rewriting:

$$-(q_1 - 12)^2 + 144 - (q_2 - 15)^2 + 225 > 269.$$

$$(q_1 - 12)^2 + (q_2 - 15)^2 < 100.$$

This represents a circle centered at $(12, 15)$ with radius 10. The inequality tells us that the profitable region consists of all points inside this circle.

Step 4: Visualize the Solution



Answer: The profitable region is the interior of the circle centered at $(12, 15)$ with radius 10, constrained by $0 \leq q_1 \leq 30$ and $0 \leq q_2 \leq 35$.

? Question 14

The produced output Q of a certain good depends on the quantities q_1 and q_2 used of two kinds of raw materials. Suppose that the dependence is given by the *production function*

$$Q = 2q_1^{2/3} q_2^{1/3}.$$

- (a) Find the equation of the isoproduction curve or *isoquant* (contour line of a production function) that contains the point $(12, 1.5)$.
- (b) Rewrite this equation into an equation that gives q_2 explicitly as a function of q_1 .
- (c) Make a sketch of the isoquant in part (a).
- (d) Find the point on the isoquant for which the value of q_2 equals 16.

💡 Tip: Understanding the Production Function

A production function represents the relationship between input quantities and the output produced. In this case, the function is given as:

$$Q = 2q_1^{2/3} q_2^{1/3}$$

which describes how the output Q depends on the raw material inputs q_1 and q_2 .

💡 Tip: Understanding Isoquants

An **isoquant** (or isoproduction curve) represents all combinations of inputs (q_1, q_2) that yield the same level of output. It is the contour line of a production function Q , just like contour lines in previous exercises.

✓ **Solution 14: (a) Isoquant of $Q = 2q_1^{2/3}q_2^{1/3}$ through (12, 1.5)**

We are given that the isoquant must pass through the point $(q_1, q_2) = (12, 1.5)$. Substituting these values into the production function $Q(q_1, q_2)$:

$$Q(12, 1.5) = 2(12)^{2/3}(1.5)^{1/3} = 12.$$

Since the isoquant represents all input combinations that produce the same output level, we generalize this equation by replacing 12 and 1.5 with variables q_1 and q_2 :

$$2 \cdot q_1^{2/3} q_2^{1/3} = 12.$$

This equation defines the isoquant, meaning all points (q_1, q_2) that satisfy this equation result in the same production output (in this case, 12).

Answer: $2 \cdot q_1^{2/3} q_2^{1/3} = 12.$

✓ **Solution 14: (b) Express q_2 as a function of q_1 along the isoquant**

Rewriting the equation:

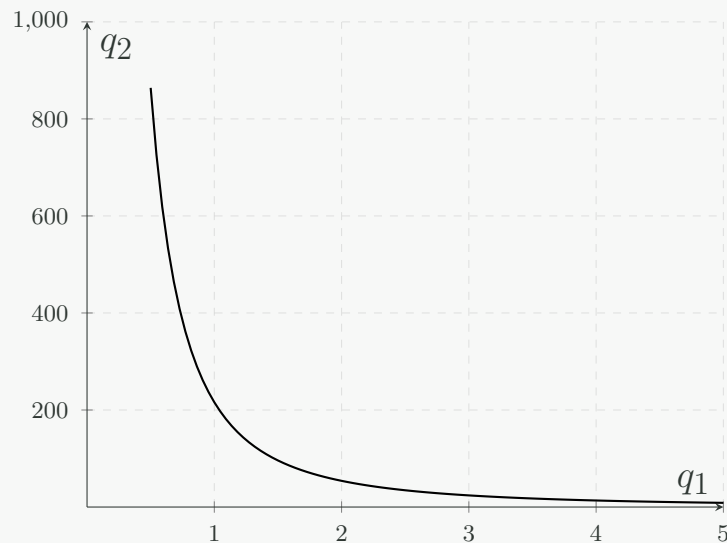
$$2 \cdot q_1^{2/3} q_2^{1/3} = 12 \Rightarrow q_2^{1/3} = \frac{6}{2 \cdot q_1^{2/3}} \Rightarrow q_2 = \frac{216}{q_1^2}.$$

Answer: $q_2 = \frac{216}{q_1^2}.$

✓ **Solution 14: (c) Graph of the isoquant $q_2 = \frac{216}{q_1^2}$**

The function $q_2 = \frac{216}{q_1^2}$ represents the isoquant passing through (12, 1.5).

Answer: The graph below illustrates this relationship.



✓ **Solution 14: (d) Finding the point on the isoquant where $q_2 = 16$**

Solving:

$$16 = \frac{216}{q_1^2} \Rightarrow q_1 = \frac{\sqrt{27}}{2} \approx 3.6742.$$

Answer: The corresponding point on the isoquant is: (3.6742, 16).

Question 15

The production q (in units) of a certain good in terms of the amount q_L of labour invested and the amount q_C of capital invested is given by the production function:

$$q = 2 \cdot q_L^{0.5} \cdot q_C^{0.5}.$$

At this moment, 6 units of labour and 6 units of capital are invested daily. By the production function, this results in 12 units being produced daily. The graphs below both show the contour line of level 12 of the production function q .

- (a) Give an implicit and explicit equation of this contour line.

Assume for this and the later parts of the question that a unit of labour costs €84 and a unit of capital costs €35. Moreover, there is a fixed cost of €486 per day.

- (b) Give the equation of the cost function $c(q_L, q_C)$.
- (c) Graph, in Figure a, a line that allows you to decide that there exists another combination (q_L, q_C) to generate 12 units a day with exactly the same cost as the combination $(q_L, q_C) = (6, 6)$.
- (d) Explain how you can approximately read this second combination from the graph above.
- (e) Calculate the second combination mentioned in part (d) exactly.
- (f) Describe the method to find the minimal budget to reach the daily production level $q = 12$ geometrically.
- (g) Calculate the minimal cost to realize production level $q = 12$ and the number of units of capital and labour that have to be invested.
- (h) Suppose the company can sell the good at €100 per unit. Explain on Figure b which combinations (q_L, q_C) to generate 12 units a day are economically sound.

Note: Refer to the original exercise document for graphs.

✓ Solution 15: (a) Implicit and explicit equations of the contour line of level 12

The contour line of level 12 satisfies the equation by substituting $q = 12$ into the production function:

$$12 = 2q_L^{0.5}q_C^{0.5}.$$

This is the implicit equation. To obtain the explicit equation, we solve for q_C :

$$q_C = \frac{36}{q_L}.$$

Answer: The implicit equation is $12 = 2q_L^{0.5}q_C^{0.5}$, and the explicit equation is $q_C = \frac{36}{q_L}$.

✓ Solution 15: (b) Cost function $c(q_L, q_C)$

First, we list all constants provided in the question:

- Cost per unit of labour: $c_L = 84$.
- Cost per unit of capital: $c_C = 35$.
- Fixed cost: $C_{\text{fixed}} = 486$.

Using the given values, the cost function becomes:

$$c(q_L, q_C) = 84q_L + 35q_C + 486.$$

Answer: $c(q_L, q_C) = 84q_L + 35q_C + 486$.

✓ Solution 15: (c) Cost contour line through $(q_L, q_C) = (6, 6)$

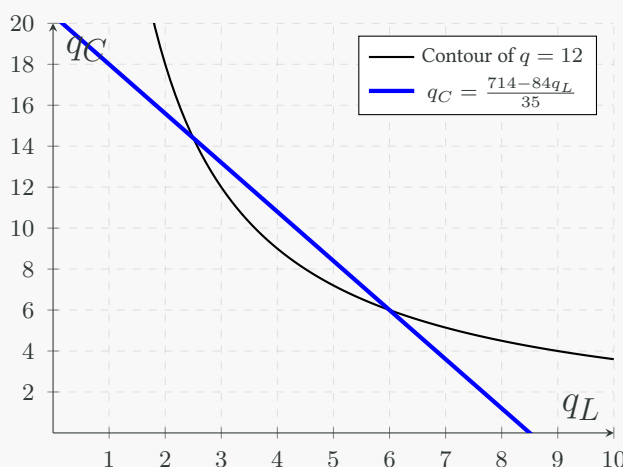
First, we calculate the total cost at $(q_L, q_C) = (6, 6)$ using the cost function from part (b):

$$c(6, 6) = 84(6) + 35(6) + 486 = 1200.$$

This means that all points with the same total cost of 1200 lie on the same cost contour line. To find the equation of that contour line, we set the cost function equal to 1200:

$$84q_L + 35q_C + 486 = 1200 \Rightarrow q_C = \frac{714 - 84q_L}{35}.$$

Answer: The equation $q_C = \frac{714 - 84q_L}{35}$ represents the contour line for a cost level of 1200. This line, shown in blue in the figure below, connects all combinations of (q_L, q_C) that result in the same total cost.



✓ Solution 15: (d) Reading the second combination from the graph

In part (c), we found the cost contour line passing through $(q_L, q_C) = (6, 6)$. This means any other point on this line represents another combination of (q_L, q_C) that results in the same total cost.

Since the production function remains fixed at 12 units, we are looking for another pair (q_L, q_C) that satisfies both the isoquant equation and the cost contour equation. In other words, we need

to find the second intersection point of these two curves.

Answer: The second combination is the intersection point (other than $(6, 6)$) between the isoquant $q_C = \frac{36}{q_L}$ and the straight-line cost function from part (c).

✓ Solution 15: (e) Exact calculation of the second combination

We set the cost function equation from part (c) equal to the explicit isoquant equation from part (a):

$$\frac{714 - 84q_L}{35} = \frac{36}{q_L} \Rightarrow 84q_L^2 - 714q_L + 1260 = 0.$$

Solving this quadratic equation, we find:

$$q_L = 6 \quad \text{or} \quad q_L \approx 2.5.$$

Since $(6, 6)$ is already known, we take the second value $q_L = 2.5$ and substitute it into the isoquant equation in order to find q_C :

$$q_C = \frac{36}{2.5} = 14.4.$$

Answer: The second combination is $(q_L, q_C) = (2.5, 14.4)$.

✓ Solution 15: (f) Finding the minimal budget geometrically

Since all contour lines of the cost function are parallel, the optimal point occurs where the isoquant has the same slope as the cost contour line from part (c). This guarantees that the cost is minimized while still producing 12 units.

Answer: The minimum cost occurs where the slope of the isoquant matches the slope of the cost contour line, ensuring the lowest possible cost for producing 12 units.

✓ Solution 15: (g) Calculating the minimal cost to realize $q = 12$

In part (f), we found that the minimal cost occurs where the isoquant is tangent to a cost contour line. This means the slopes of both functions are equal.

If we look at the equation of the contour line of the cost function that we found in part (c), they all have a slope of $-\frac{84}{35}$. To find the point where the isoquant has this same slope, we solve for q_L by equating the derivative of the isoquant $q_C = \frac{36}{q_L}$ to this slope:

$$\frac{d}{dq_L} \left(\frac{36}{q_L} \right) = -\frac{84}{35} \Rightarrow -\frac{36}{q_L^2} = -\frac{84}{35} \Rightarrow q_L = \sqrt{15}$$

Hence, we find:

$$(q_L, q_C) = \left(\sqrt{15}, \frac{36}{\sqrt{15}} \right) \approx (3.87, 9.29).$$

Now, we calculate the cost at this point using the cost function:

$$c(q_L, q_C) = 84q_L + 35q_C + 486 \Rightarrow c(3.87, 9.29) = 1136.66$$

Answer: The minimal cost to produce 12 units is approximately €1136.66.

✓ Solution 15: (h) Economically sound combinations at €100 per unit

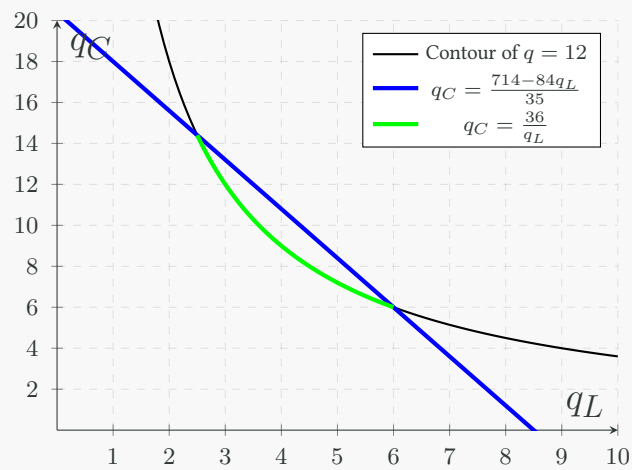
The company sells each unit at €100, so the total revenue for producing 12 units is:

$$R = 100 \cdot 12 = 1200.$$

For a combination to be economically viable, its total cost must be below this revenue threshold. The contour line of cost level 1200 defines this limit. Any point on the isoquant below this line represents a combination where production remains profitable.

Answer: The economically sound combinations are the points on the isoquant $q_C = \frac{36}{q_L}$ that lie below the cost contour line of level 1200. These points are highlighted in green in the figure

below.



Question 16

The production q (in units) of a certain good in terms of the amount q_L of labour invested and the amount q_C of capital invested is given by the production function:

$$q = 5 \cdot q_L^{2/3} \cdot q_C^{1/3}.$$

At this moment, 512 units of labour and 1000 units of capital are invested daily. By the production function, this results in 3200 units being produced daily. The graph below shows the contour line of level 3200 of the production function q .

(a) Give an implicit and explicit equation of this contour line.

Assume for this and the later parts of the question that a unit of labour costs 1.89 units of money and that a unit of capital costs 1.25 units of money. Moreover, there is a fixed cost of 14.3 units of money per day.

(b) Give the equation of the cost function $c(q_L, q_C)$.

(c) Graph, in Figure below, a line that allows you to decide that there exists another combination (q_L, q_C) to generate 3200 units a day with exactly the same cost as the combination $(q_L, q_C) = (512, 1000)$.

(d) Explain how you can approximately read this second combination from the graph above.

(e) Calculate the second combination mentioned in part (d) exactly.

(f) Explain how you can graphically find the combinations to produce 3200 units a day in a cheaper way than by investing $(q_L, q_C) = (512, 1000)$.

(g) Describe the method to find the minimal budget to reach the daily production level $q = 3200$ geometrically.

(h) Calculate the minimal cost to realize production level $q = 3200$ and the number of units of capital and labour that have to be invested.

Note: Refer to the original exercise document for graphs.

💡 Tip: Similarity to Previous Question

This question follows the same structure as the previous one, with different numerical values. The steps to solve it are the same.

Only part of the question that is different is (f).

✓ Solution 16: (a) Implicit and explicit equations of the contour line of level 3200

The contour line of level 3200 satisfies the equation by substituting $q = 3200$ into the production function:

$$3200 = 5q_L^{2/3} q_C^{1/3}.$$

This is the implicit equation. To obtain the explicit equation, we solve for q_C :

$$q_C = \frac{640}{q_L^2}.$$

Answer: The implicit equation is $3200 = 5q_L^{2/3} q_C^{1/3}$, and the explicit equation is $q_C = \frac{640}{q_L^2}$.

✓ Solution 16: (b) Cost function $c(q_L, q_C)$

First, we list all constants provided in the question:

- Cost per unit of labour: $c_L = 1.89$.
- Cost per unit of capital: $c_C = 1.25$.
- Fixed cost: $C_{\text{fixed}} = 14.3$.

Using the given values, the cost function becomes:

$$c(q_L, q_C) = 1.89q_L + 1.25q_C + 14.3.$$

Answer: $c(q_L, q_C) = 1.89q_L + 1.25q_C + 14.3$.

✓ **Solution 16: (c) Cost contour line through $(q_L, q_C) = (512, 1000)$**

First, we calculate the total cost at $(q_L, q_C) = (512, 1000)$ using the cost function from part (b):

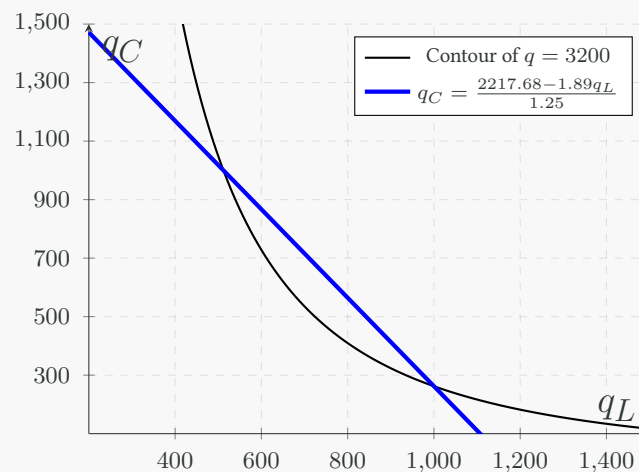
$$c(512, 1000) = 1.89(512) + 1.25(1000) + 14.3 = 2231.98.$$

This means that all points with the same total cost of 2231.98 lie on the same cost contour line.

To find the equation of that contour line, we set the cost function equal to 2231.98:

$$1.89q_L + 1.25q_C + 14.3 = 2231.98 \Rightarrow q_C = \frac{2217.68 - 1.89q_L}{1.25}.$$

Answer: The equation $q_C = \frac{2217.68 - 1.89q_L}{1.25}$ represents the contour line for a cost level of 2231.98. This line, shown in blue in the figure below, connects all combinations of (q_L, q_C) that result in the same total cost.



✓ Solution 16: (d) Reading the second combination from the graph

In part (c), we found the cost contour line passing through $(q_L, q_C) = (512, 1000)$. This means any other point on this line represents another combination of (q_L, q_C) that results in the same total cost.

Since the production function remains fixed at 3200 units, we are looking for another pair (q_L, q_C) that satisfies both the isoquant equation and the cost contour equation. In other words, we need to find the second intersection point of these two curves.

Answer: The second combination is the intersection point (other than $(512, 1000)$) between the isoquant $q_C = \frac{640^3}{q_L^2}$ and the straight-line cost function from part (c).

💡 Tip: Solving for the Intersection

To find the intersection of two curves, set their equations equal to each other, solve for q_L , and then use the result to find q_C .

✓ Solution 16: (e) Exact calculation of the second combination

We set the cost function equation from part (c) equal to the explicit isoquant equation from part (a):

$$\frac{2217.68 - 1.89q_L}{1.25} = \frac{640^3}{q_L^2}.$$

Solving for q_L , we find:

$$q_L^1 \approx 338.62, \quad q_L^1 = 512, \quad q_L^3 = 1000$$

Since $q_L^1 = 512$ is negative (which is impossible as this is quantity), and $q_L^2 = 512$ is already known, we take the last value $q_L = 1000$ and substitute it into the isoquant equation to find q_C :

$$q_C = \frac{640^3}{1000^2} = 262.144.$$

Answer: The second combination is $(q_L, q_C) = (1000, 262.144)$.

Tip: Lower Cost Regions

If a point on the isoquant lies below the straight-line cost contour, then the cost of that combination is lower because the contour lines of the cost function indicate cost levels. Points below a given contour line correspond to lower cost levels.

Solution 16: (f) Finding cheaper combinations graphically

Answer: Graphically, the valid cheaper combinations are the points on the isoquant $q_C = \frac{640^3}{q_L^2}$ that are below the cost contour line equation from part (c). This is because cost contour lines represent equal cost levels, and any point below the given contour level corresponds to a lower cost.

Tip: Isoquant Tangency Condition

The minimal cost occurs where the isoquant is tangent to a cost contour line. This means the slopes of both curves are equal, ensuring the lowest possible cost for the given production level.

Solution 16: (g) Finding the minimal budget geometrically

Since all contour lines of the cost function are parallel, the optimal point occurs where the isoquant has the same slope as the cost contour line from part (c). This guarantees that the cost is minimized while still producing 3200 units.

Answer: The minimum cost occurs where the slope of the isoquant matches the slope of the cost contour line, ensuring the lowest possible cost for producing 3200 units.

💡 Tip: Understanding Tangent Slopes

The slope of a function at any given point is given by its derivative.

✓ Solution 16: (h) Calculating the minimal cost to realize $q = 3200$

In part (f), we found that the minimal cost occurs where the isoquant is tangent to a cost contour line. This means the slopes of both functions are equal.

If we look at the equation of the contour line of the cost function that we found in part (c), they all have a slope of $-\frac{1.89}{1.25}$. To find the point where the isoquant has this same slope, we solve for q_L by equating the derivative of the isoquant $q_C = \frac{640^3}{q_L^2}$ to this slope:

$$\frac{d}{dq_L} \left(\frac{640^3}{q_L^2} \right) = -\frac{2 \cdot 640^3}{q_L^3} = -\frac{1.89}{1.25}.$$

Solving for q_L , we find:

$$q_L \approx 702.54$$

Substituting into the explicit equation obtained in part (a), we get:

$$q_C = \frac{640}{702.54^2} = 531.12.$$

Now, we calculate the cost at this point using the cost function:

$$c(q_L, q_C) = 1.89q_L + 1.25q_C + 14.3 \Rightarrow c(702.54, 531.12) \approx 2006$$

Answer: The minimal cost to produce 3200 units is approximately €2006.

Question 17

The production q (in units) of a certain good is given by the production function:

$$q = q_L^2 q_C,$$

where q represents the output produced in units of 1000, and q_L and q_C are the number of units of labour and capital invested in the production process.

Assume that:

- One unit of labour costs 10 units of money.
 - One unit of capital costs 5 units of money.
- (a) Show on a graph which combinations (q_L, q_C) generate exactly 64 000 units of output. Put q_L on the horizontal axis of the graph and q_C on the vertical axis.
 - (b) Explain how one can see on the graph of part (a) which combinations (q_L, q_C) allow production of 64 000 units using a budget of exactly 100 units of money.
 - (c) Explain how one can see on the graph of part (a) which combinations (q_L, q_C) allow production of 64 000 units in the cheapest possible way.

✓ Solution 17: (a) Graph of the isoquant for 64,000 units of $q = q_L^2 q_C$

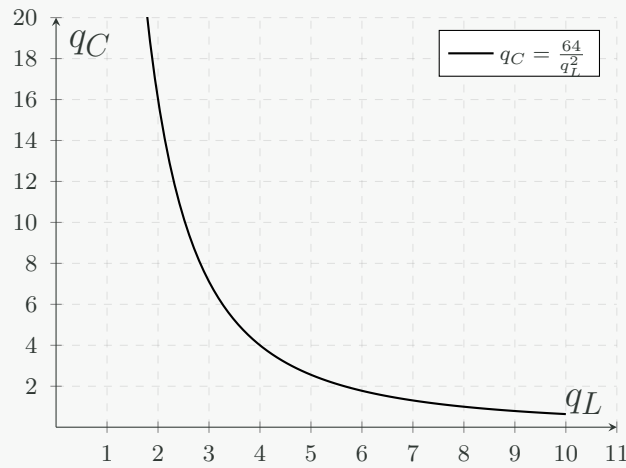
The production function is given as:

$$q = q_L^2 q_C.$$

Since q is measured in thousands, we substitute $q = 64$ to obtain the equation of the isoquant:

$$64 = q_L^2 q_C \quad \Rightarrow \quad q_C = \frac{64}{q_L^2}.$$

Answer: The equation $q_C = \frac{64}{q_L^2}$ defines the contour line of level 64 of the production function. The graph is shown below:



✓ Solution 17: (b) Combinations for a budget of 100 units of money

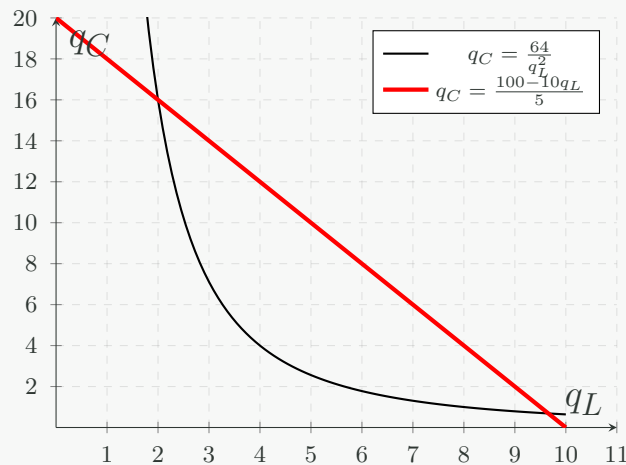
The cost function is given as:

$$c(q_L, q_C) = 10q_L + 5q_C.$$

For a total budget of 100, we set:

$$10q_L + 5q_C = 100 \quad \Rightarrow \quad q_C = \frac{100 - 10q_L}{5}.$$

Answer: The valid combinations of (q_L, q_C) are the intersection points of this line with the isoquant from part (a), as shown below:



✓ Solution 17: (c) Finding the cheapest production combination

To minimize cost, we need to find the contour line of the cost function that is tangent to the isoquant from part (a). Since all contour lines of $c(q_L, q_C)$ are parallel, the optimal solution occurs where the isoquant has the same slope as one of these contour lines.

The contour lines of the cost function are given by:

$$c(q_L, q_C) = 10q_L + 5q_C.$$

Since all cost contour lines are parallel, they share the same slope:

$$\frac{dq_C}{dq_L} = -\frac{10}{5} = -2.$$

To find the point where the isoquant is tangent to this cost contour, we differentiate the isoquant equation:

$$q_C = \frac{64}{q_L^2}.$$

Taking the derivative:

$$\frac{dq_C}{dq_L} = -\frac{128}{q_L^3}.$$

Setting this equal to the slope of the cost function:

$$-\frac{128}{q_L^3} = -2 \Rightarrow q_L = 4.$$

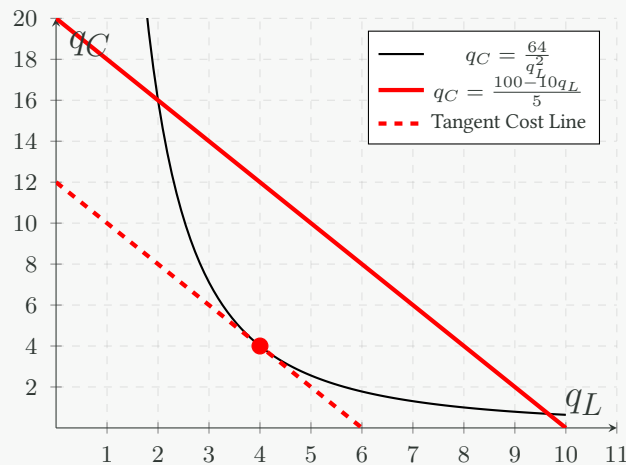
Substituting $q_L = 4$ into the isoquant equation:

$$q_C = \frac{64}{4^2} = \frac{64}{16} = 4.$$

Thus, the optimal point is:

$$(q_L, q_C) = (4, 4).$$

This corresponds to the dashed red line in the figure below, which represents the cost contour line tangent to the isoquant.



Answer: The cheapest way to produce 64,000 units occurs at $(q_L, q_C) = (4, 4)$, where the isoquant and cost contour line are tangent.

? Question 18

The cost c (in €) for the production of q_1 units of good 1 and q_2 units of good 2 is:

$$c(q_1, q_2) = 6q_1 + 4q_2 + 525.$$

The demand functions for good 1 and good 2 are given by:

$$p_1 = 36 - q_1, \quad p_2 = 44 - q_2.$$

- (a) Find the equation of the profit function.
- (b) Show on a graph which combinations (q_1, q_2) give neither profit nor loss.
- (c) Currently, 9 units of the first good and 24 units of the second good are being produced, and the firm makes a profit in this way.
 - i. Is it possible to realize the same profit on a lower budget? Use contour lines for profit and cost to answer this question graphically.
 - ii. Estimate which production amounts lead to the *lowest* budget without affecting the profit made from the graph. Also, estimate the lowest budget.
 - iii. Now find exact values for the production amounts and the lowest budget by computation.
- (d) The firm has a budget of 669 monetary units, which it spends completely.
 - i. Show the corresponding isocost line graphically.
 - ii. Add a contour plot for the profit function and use this to estimate the maximum profit that the company can realize with this budget and the corresponding production amounts.
 - iii. Now find the exact values for the production amounts and the maximum profit by computation.

✓ Solution 18: (a) Finding the profit function

Profit is total revenue minus cost, so we start with:

$$P(q_1, q_2) = p_1 \cdot q_1 + p_2 \cdot q_2 - c(q_1, q_2).$$

Substituting the given demand and cost functions:

$$P(q_1, q_2) = (36 - q_1)q_1 + (44 - q_2)q_2 - (6q_1 + 4q_2 + 525).$$

Simplifying:

$$P(q_1, q_2) = -q_1^2 - q_2^2 + 30q_1 + 40q_2 - 525.$$

Answer: The profit function is:

$$P(q_1, q_2) = -q_1^2 - q_2^2 + 30q_1 + 40q_2 - 525.$$

✓ Solution 18: (b) Combinations giving neither profit nor loss

The problem requires us to determine which combinations (q_1, q_2) yield neither profit nor loss, meaning we solve:

$$P(q_1, q_2) = 0.$$

Rewriting the profit function:

$$-q_1^2 - q_2^2 + 30q_1 + 40q_2 - 525 = 0.$$

We complete the square for both q_1 and q_2 :

$$-(q_1^2 - 30q_1) - (q_2^2 - 40q_2) = 525.$$

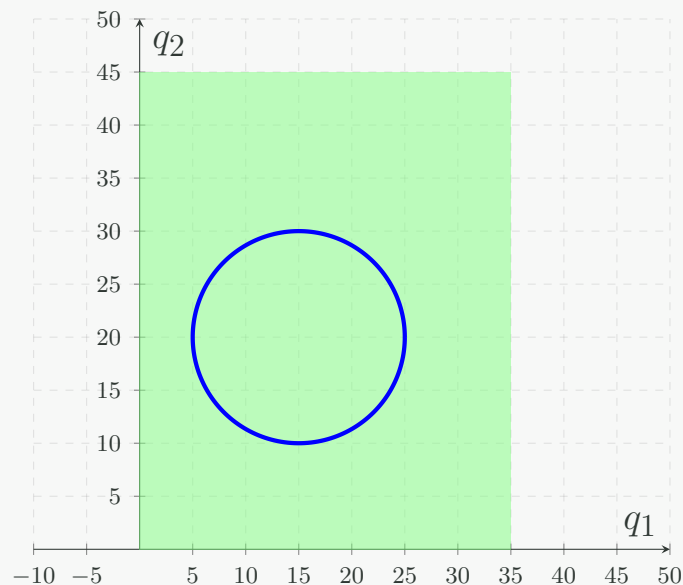
Rewriting,

$$-(q_1 - 15)^2 + 225 - (q_2 - 20)^2 + 400 = 525.$$

$$(q_1 - 15)^2 + (q_2 - 20)^2 = 100.$$

This equation represents a circle centered at $(15, 20)$ with radius 10. The points on this circle line correspond to combinations where the company makes neither profit nor loss.

On the graph below, the green region represents the valid domain, while the blue circle indicates the zero-profit combinations.



Answer: The equation $(q_1 - 15)^2 + (q_2 - 20)^2 = 100$ defines the set of combinations where profit is zero. This region is highlighted in the graph below.

✓ Solution 18: (c.i) Using contour lines to analyze profit and cost at (9, 24)

To draw the contour lines for profit and cost, we first determine the current profit level. Since the company produces at $(q_1, q_2) = (9, 24)$, we calculate:

$$P(9, 24) = -9^2 - 24^2 + 30(9) + 40(24) - 525 = 48.$$

Thus, the contour line for profit level 48 satisfies:

$$-q_1^2 - q_2^2 + 30q_1 + 40q_2 - 525 = 48.$$

Step 1: Rewrite as a Circle Equation

Rearranging:

$$-q_1^2 - q_2^2 + 30q_1 + 40q_2 = 573.$$

Completing the square:

$$-(q_1^2 - 30q_1) - (q_2^2 - 40q_2) = 573.$$

$$-(q_1 - 15)^2 + 225 - (q_2 - 20)^2 + 400 = 573.$$

$$(q_1 - 15)^2 + (q_2 - 20)^2 = 52.$$

This represents a circle centered at (15, 20) with radius $\sqrt{52}$.

Step 2: Cost Contour Line

The general equation for the cost contour lines is:

$$6q_1 + 4q_2 + 525 = a \quad \Rightarrow \quad q_2 = \frac{-6q_1 + (a - 525)}{4}.$$

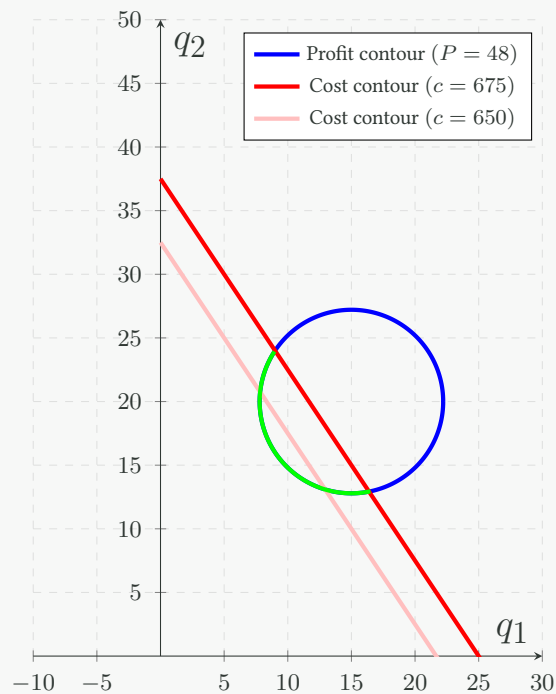
Since all cost contour lines share the slope:

$$\frac{dq_2}{dq_1} = -\frac{6}{4} = -\frac{3}{2},$$

they are parallel lines with a slope of $-3/2$. For example, for $a = 675$, the contour line equation becomes:

$$q_2 = \frac{675 - 6q_1 - 525}{4} = \frac{150 - 6q_1}{4}.$$

We see that all points on the arc below this contour remain at profit level 48 (meaning profit remains unchanged) but correspond to a lower cost contour than the current level (for example, 650).



Answer: Moving along the arc below this cost contour allows the company to maintain the same profit level while reducing costs.

✓ Solution 18: (c.ii) Estimating the lowest budget for the same profit

To find the production amounts that achieve the lowest budget while maintaining the same profit level, we need to lower the cost contour while staying on the profit contour for $P = 48$. This means shifting the current cost contour downward in a parallel manner until it touches the profit contour at exactly one point.

The very last point on the profit contour before the cost contour moves below it represents the lowest budget needed to maintain the same profit. A good estimate for this point is:

$$(q_1, q_2) = (9, 16),$$

which corresponds to a cost of 645.

Answer: The estimated lowest budget while maintaining $P = 48$ is 645 at $(q_1, q_2) = (9, 16)$.

✓ Solution 18: (c.iii) Exact calculation of the lowest budget production point

To determine the exact production amounts that achieve the lowest budget while maintaining the same profit, we must find the point where the cost contour line is tangent to the profit contour at level $P = 48$.

Since all cost contour lines have a slope of:

$$\frac{dq_2}{dq_1} = -\frac{3}{2},$$

we need to find where the profit contour has the same slope.

Step 1: Expressing the Profit Contour Explicitly

The contour line for profit level 48 is:

$$(q_1 - 15)^2 + (q_2 - 20)^2 = 52.$$

Solving for q_2 :

$$q_2 = 20 \pm \sqrt{52 - (q_1 - 15)^2}.$$

Since we are minimizing cost, we take the negative root to remain on the bottom half of the circle.

$$q_2 = 20 - \sqrt{52 - (q_1 - 15)^2}.$$

Step 2: Solving for q_1

Differentiating both sides:

$$\frac{dq_2}{dq_1} = \frac{q_1 - 15}{\sqrt{52 - (q_1 - 15)^2}}.$$

Setting this derivative equal to the slope of the cost contour line:

$$\frac{q_1 - 15}{\sqrt{52 - (q_1 - 15)^2}} = -\frac{3}{2} \Rightarrow q_1 = 9$$

Substituting into the profit contour equation to solve for q_2 :

$$(9 - 15)^2 + (q_2 - 20)^2 = 49 \Rightarrow q_2 \approx 16$$

Answer: The exact production amounts that achieve the lowest budget while maintaining $P = 48$ are $(q_1, q_2) = (9, 16)$.

✓ Solution 18: (d.i) Isocost line for a budget of 669

The firm has a total budget of 669. The cost function is given as:

$$c(q_1, q_2) = 6q_1 + 4q_2 + 525.$$

Setting this equal to 669:

$$6q_1 + 4q_2 + 525 = 669.$$

Solving for q_2 :

$$q_2 = \frac{669 - 6q_1 - 525}{4} = \frac{144 - 6q_1}{4} = 36 - \frac{3}{2}q_1.$$

Answer: The isocost line for a budget of 669 is given by $q_2 = 36 - \frac{3}{2}q_1$.

✓ Solution 18: (d.ii) Estimating the maximum profit with a budget of 669

To determine the highest possible profit while maintaining a total budget of 669, we analyze the contour lines of the profit function and their intersection with the isocost line.

Since the profit function is given by:

$$P(q_1, q_2) = -q_1^2 - q_2^2 + 30q_1 + 40q_2 - 525,$$

we rewrite the contour equation for any profit level a :

$$(q_1 - 15)^2 + (q_2 - 20)^2 = 100 - a.$$

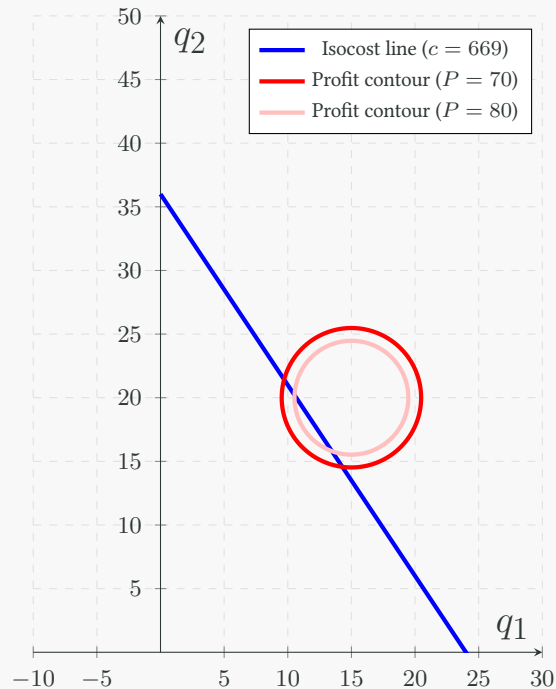
To maximize profit, we seek the highest profit contour that touches the isocost line:

$$q_2 = 36 - \frac{3}{2}q_1.$$

By gradually decreasing the radius of the profit contour, we find that the last point of intersection before the contour moves inside the isocost line occurs at approximately:

$$(q_1, q_2) = (12, 18),$$

corresponding to a profit of 87.



Answer: The estimated maximum profit attainable with a budget of 669 is 87, at $(q_1, q_2) = (12, 18)$.

✓ Solution 18: (d.iii) Exact calculation of the maximum profit production point

We now use a geometric approach instead of derivatives. We find the point of tangency between the profit contour (circle) and the isocost line using perpendicularity.

The profit contours are circles centered at:

$$(15, 20)$$

The isocost line was previously found as:

$$q_2 = 36 - \frac{3}{2}q_1$$

So its slope is $m = -\frac{3}{2}$. A perpendicular line must have slope:

$$m_{\perp} = \frac{2}{3}$$

We now write the line through $(15, 20)$ with slope $\frac{2}{3}$:

$$q_2 - 20 = \frac{2}{3}(q_1 - 15) \Rightarrow q_2 = \frac{2}{3}q_1 + 10$$

We now solve for the point where this perpendicular line intersects the isocost line $q_2 = 36 - \frac{3}{2}q_1$:

$$\frac{2}{3}q_1 + 10 = 36 - \frac{3}{2}q_1$$

Multiply both sides by 6 to eliminate denominators:

$$4q_1 + 60 = 216 - 9q_1 \Rightarrow 13q_1 = 156 \Rightarrow q_1 = 12$$

Substitute back into either equation to find q_2 :

$$q_2 = \frac{2}{3} \cdot 12 + 10 = 8 + 10 = 18$$

Answer: The exact production point that maximizes profit under the isocost line of 669 is:

$$(q_1, q_2) = (12, 18)$$

Profit in this case is $P(12, 18) = 87$.

? Question 19

A firm produces two products:

- A **niche product** with an inverse demand function given by:

$$p_1 = 100 - 2q_1.$$

- A **standard product** with a fixed price of 2 monetary units, meaning:

$$p_2 = 2.$$

The variable production cost for the niche product is 5 monetary units per unit of production, while for the standard product it is 1 monetary unit per unit of production. There is also a fixed production cost of 100 monetary units.

- Find the equation of the revenue function $R(q_1, q_2)$, as well as its domain.
- Show on a graph which production amounts for the two products lead to a revenue strictly between 1000 and 1500 monetary units.
- The firm has a budget of 400 monetary units. Add the corresponding isocost line to the graph from part b.
- Indicate on the graph the production amounts for the two products that lead to the budget of 400 monetary units being fully spent, while generating revenue between 1000 and 1500 monetary units.
- What is the minimal budget needed for a revenue of 1000 monetary units? What is the minimal budget needed for a revenue of 1500 monetary units? First answer this question graphically (giving an approximate answer), and then exactly by calculation.

✓ Solution 19: (a) Finding the revenue function $R(q_1, q_2)$ and its domain

The revenue function is defined as total revenue from both products:

$$R(q_1, q_2) = p_1 \cdot q_1 + p_2 \cdot q_2.$$

Substituting the given inverse demand functions:

$$R(q_1, q_2) = (100 - 2q_1)q_1 + 2q_2.$$

Expanding:

$$R(q_1, q_2) = 100q_1 - 2q_1^2 + 2q_2.$$

Answer: The revenue function is:

$$R(q_1, q_2) = 100q_1 - 2q_1^2 + 2q_2,$$

with domain:

$$0 \leq q_1 \leq 50, \quad 0 \leq q_2.$$

✓ Solution 19: (b) Production amounts giving revenue strictly between 1000 and 1500

The revenue function found in part (a) defines a parabolic curve in the (q_1, q_2) -plane. To analyze which production amounts lead to revenues between 1000 and 1500, we plot the contour lines for $R = 1000$ and $R = 1500$.

Note: Refer to the original solution document for graphs.

Answer: The region between two parabolic curves highlights production amounts that result in revenue values between 1000 and 1500.

✓ Solution 19: (c) Isocost line for a budget of 400 monetary units

To determine where the revenue satisfies $1000 < R(q_1, q_2) < 1500$, we set up the general equation for the revenue contour line at level a :

$$R(q_1, q_2) = a \Rightarrow q_2 = q_1^2 - 50q_1 + \frac{a}{2}.$$

This formula helps locate the maximum or minimum point of the parabola. In our case, using $A = 1$ and $B = -50$, we find:

$$q_1 = \frac{50}{2} = 25, \quad q_2 = \frac{a - 1250}{2}.$$

Additionally, the cost function is given by:

$$c(q_1, q_2) = 5q_1 + q_2 + 100.$$

The isocost line for cost level 400 is found by setting $c(q_1, q_2) = 400$:

$$q_2 = 300 - 5q_1.$$

Note: Refer to the original solution document for graphs.

Answer: The green region represents the valid production amounts where revenue is between 1000 and 1500. The blue/brown isocost line for $c = 400$ shows the cost constraint.

✓ Solution 19: (d) Production amounts spending budget of 400 with revenue between 1000 and 1500

Answer: Refer to the original solution document for graphs.

✓ Solution 19: (e) Minimal budget for revenues of 1000 and 1500 monetary units

The general equation for an isocost line at level a is:

$$q_2 = (a - 100) - 5q_1.$$

All isocost lines share the same slope of -5 , meaning they are parallel. Lower cost levels correspond to a lower vertical intercept. We now determine the minimum cost needed to achieve revenue levels of 1000 and 1500.

Step 1: Finding the Minimal Cost for $R = 1000$

To find the lowest cost required to achieve $R = 1000$, we lower the isocost line in a parallel way until the very last point where it intersects the revenue contour at level 1000.

The equation for the revenue contour at $R = 1000$ is:

$$q_2 = q_1^2 - 50q_1 + 500.$$

To find the minimum budget, we solve for the zeroes:

$$0 = q_1^2 - 50q_1 + 500 \Rightarrow q_1 \approx 13.82.$$

Substituting into the cost function:

$$c = 5(13.82) + 0 + 100 = 169.10.$$

Step 2: Finding the Minimal Cost for $R = 1500$

Similarly, to achieve $R = 1500$ with the lowest possible budget, we move the isocost line downward until it is tangent to the revenue contour at $R = 1500$.

The revenue contour at $R = 1500$ is:

$$q_2 = q_1^2 - 50q_1 + 750.$$

The minimum budget occurs where the isocost line is tangent to the revenue contour, meaning their slopes must be equal. The slope of the isocost lines is always -5 , so we solve:

$$\frac{dq_2}{dq_1} = 2q_1 - 50 = -5 \Rightarrow q_1 = 22.5.$$

Substituting into the revenue equation:

$$q_2 = (22.5)^2 - 50(22.5) + 750 = 131.25.$$

Finally, the cost at this point is:

$$c = 5(22.5) + 131.25 + 100 = 343.75.$$

Answer:

- The exact minimum budget needed to achieve $R = 1000$ is 169.10 at $(q_1, q_2) \approx (13.82, 0)$.
- The exact minimum budget needed to achieve $R = 1500$ is 343.75 at $(q_1, q_2) \approx (22.5, 131.25)$.

Note: Refer to the original solution document for graphs.

Review 1 - Functions of Two Variables

Question 1

Consider the function $z = f(x, y)$ given by:

$$z = \frac{x + y}{1 + x^2 + y^2}.$$

- (a) Find the contour lines for levels $-1, -0.5, 0, 0.5$ and 1 .
- (b) Find the general contour line, i.e., for an arbitrary level a .
- (c) From (b), find the range of the function f .

Note: Refer to the original exercise document for graphs.

Tip: Steps for Finding Contour Lines

To find contour lines of a function $f(x, y)$, follow these steps:

1. **Find the domain** – Determine where $f(x, y)$ is defined.
2. **Set up the contour equation** – Solve $f(x, y) = a$ for a general constant a .
3. **Check when it exists** – Identify which values of a lead to valid contour lines.
4. **Draw the contour lines** – Plot the contour lines for the given levels.

✓ Solution 1: (a) Finding the Contour Lines for Specific Levels

Note: In this question, we could directly plug in the given contour levels and solve for explicit equations immediately. However, we will follow a systematic procedure to show a general method that applies to any question. This ensures consistency and makes it clear that contour lines can always be derived this way.

For parts (b) and (c), we will refer back to the general contour equation found in part (a), making the solution more structured and avoiding unnecessary repetition.

Step 1: Find the domain

The function given is:

$$f(x, y) = \frac{x + y}{1 + x^2 + y^2}.$$

The denominator $1 + x^2 + y^2$ is always positive for all $(x, y) \in \mathbb{R}^2$, meaning the function is well-defined everywhere. Therefore, there are no domain restrictions.

Step 2: Set up the contour equation

Setting $f(x, y) = a$ for a given level a , we solve:

$$\frac{x + y}{1 + x^2 + y^2} = a.$$

Rearranging:

$$x + y = a(1 + x^2 + y^2).$$

Rewriting:

$$ax^2 + ay^2 - x - y + a = 0.$$

Step 3: Check for existence

To determine when valid contour lines exist, we analyze different cases based on the value of a .

Case 1: $a = 0$

Setting $a = 0$ in the contour equation:

$$0 \cdot x^2 + 0 \cdot y^2 - x - y + 0 = 0 \quad \Rightarrow \quad y = -x.$$

This is the equation of a straight line passing through the origin with slope -1 .

Case 2: $a \neq 0$

We complete the square for x and y . First, rewrite:

$$ax^2 - x + ay^2 - y + a = 0.$$

As $a \neq 0$ in this case, we can factor out a from the squared terms:

$$x^2 - \frac{x}{a} + y^2 - \frac{y}{a} = -1.$$

Complete the square inside the parentheses:

$$\left(x - \frac{1}{2a}\right)^2 - \frac{1}{4a^2} + \left(y - \frac{1}{2a}\right)^2 - \frac{1}{4a^2} = -1.$$

Rearrange:

$$\left(x - \frac{1}{2a}\right)^2 + \left(y - \frac{1}{2a}\right)^2 = -1 + \frac{1}{2a^2}.$$

For this equation to represent a circle, the right-hand side must be positive:

$$-1 + \frac{1}{2a^2} > 0 \quad \Rightarrow \quad -\frac{1}{\sqrt{2}} < a < \frac{1}{\sqrt{2}}$$

Otherwise, there are no real solutions, meaning no valid contour lines exist in these cases.

$$\left\{ \begin{array}{ll} y = -x, & \text{if } a = 0 \\ \left(x - \frac{1}{2a}\right)^2 + \left(y - \frac{1}{2a}\right)^2 = -1 + \frac{1}{2a^2}, & \text{if } -\frac{1}{\sqrt{2}} < a < \frac{1}{\sqrt{2}}, a \neq 0 \\ \text{no real solution,} & \text{if } |a| > \frac{1}{\sqrt{2}}. \end{array} \right.$$

Step 4: Plot the contour lines

We now plot the contour lines for different values of a :

$$a = -1 \quad \Rightarrow \quad \text{No contour line}$$

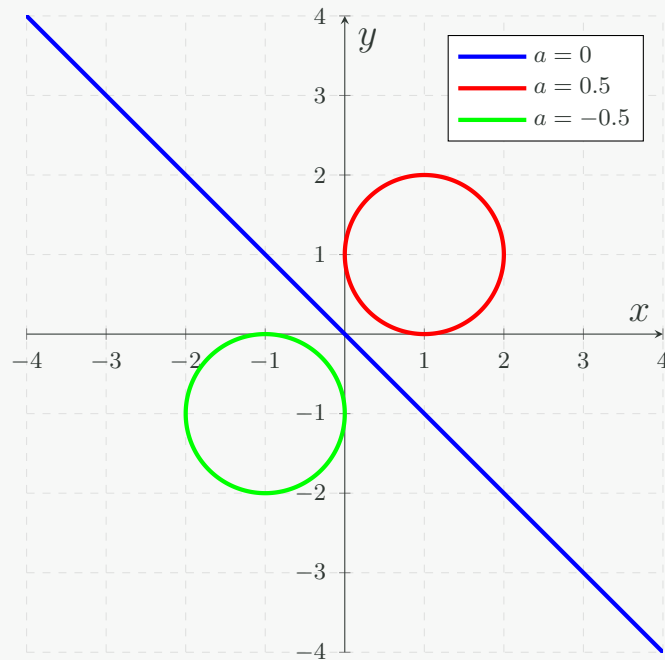
$$a = -0.5 \quad \Rightarrow \quad (x + 1)^2 + (y + 1)^2 = 1$$

$$a = 0 \quad \Rightarrow \quad y = -x$$

$$a = 0.5 \quad \Rightarrow \quad (x - 1)^2 + (y - 1)^2 = 1$$

$$a = 1 \quad \Rightarrow \quad \text{No contour line}$$

Answer:



✓ Solution 1: (b) Finding the General Contour Equation

Answer: The general contour equation for any level (a) is already derived and follows the cases outlined below.

$$\left\{ \begin{array}{ll} y = -x, & \text{if } a = 0 \\ \left(x - \frac{1}{2a}\right)^2 + \left(y - \frac{1}{2a}\right)^2 = -1 + \frac{1}{2a^2}, & \text{if } -\frac{1}{\sqrt{2}} < a < \frac{1}{\sqrt{2}}, a \neq 0 \\ \text{no real solution,} & \text{if } |a| > \frac{1}{\sqrt{2}}. \end{array} \right.$$

✓ Solution 1: (c) Finding the Range of f

Since we determined in part (b) that contour lines only exist for values satisfying:

$$-\frac{1}{\sqrt{2}} \leq a \leq \frac{1}{\sqrt{2}},$$

this directly tells us the range of $f(x, y)$, as these are the only possible output values the function can attain.

Answer: The range of $f(x, y)$ is:

$$\left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right].$$

Question 2

A company needs two types of raw materials for the production of a certain product. The production function is given by:

$$Q(q_1, q_2) = 0.2q_1q_2 + 0.04q_1^2$$

where q_1 represents the number of units of the first raw material and q_2 represents the number of units of the second raw material used in the production process. The cost of one unit of the first raw material is €6, and the cost of one unit of the second raw material is €4. Assume all variables are continuous.

- (a) The isoquant of level 300 is given. Find the explicit equation of this isoquant.
- (b) Derive the equation for the total cost function $TC(q_1, q_2)$ and sketch isocost lines for levels 200, 300, and 400.
- (c) Explain how the completed figure can be used to determine the combination of raw materials that results in the lowest cost for producing exactly 300 units. Estimate the values from the graph and explain.
- (d) Compute the exact values for the combination of raw materials that minimize the cost while producing exactly 300 units.

Note: Refer to the original exercise document for graphs.

Tip: Understanding an Isoquant

An **isoquant** represents all combinations of input quantities that produce the same level of output. To find its equation, we set the production function equal to a given output level and solve for one variable in terms of the other.

✓ Solution 2: (a) Finding the Isoquant Equation for $Q = 300$

The production function is given as:

$$Q(q_1, q_2) = 0.2q_1q_2 + 0.04q_1^2.$$

For the isoquant of level $Q = 300$, we set:

$$0.2q_1q_2 + 0.04q_1^2 = 300.$$

Rearrange the equation to express q_2 in terms of q_1 :

$$q_2 = \frac{300 - 0.04q_1^2}{0.2q_1} \Rightarrow q_2 = \frac{1500}{q_1} - 0.2q_1.$$

Answer: The equation of the isoquant for $Q = 300$ is:

$$q_2 = \frac{1500}{q_1} - 0.2q_1.$$

💡 Tip: Understanding Isocost Lines

Isocost lines represent combinations of inputs that result in the same total cost. Each isocost line corresponds to a fixed budget level.

✓ Solution 2: (b) Finding the Total Cost Function**Step 1: Deriving the Cost Function**

The total cost function is determined by summing the costs of both raw materials:

$$TC(q_1, q_2) = c_1q_1 + c_2q_2 = 6q_1 + 4q_2$$

Step 2: Finding the Specific Isocost Lines

To visualize cost levels, we set $TC(q_1, q_2)$ equal to specific values:

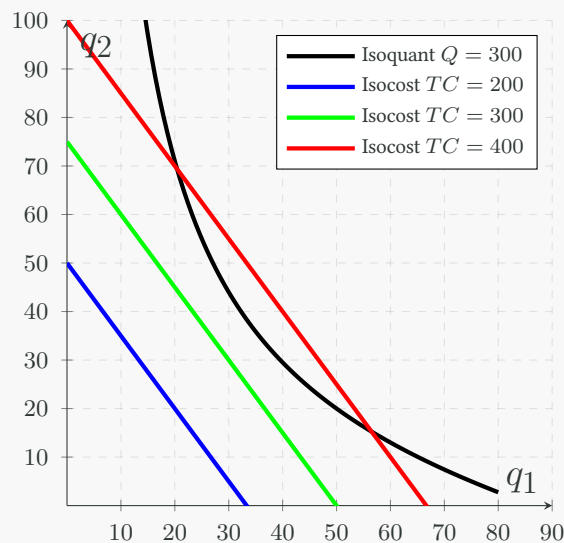
$$200 = 6q_1 + 4q_2, \quad 300 = 6q_1 + 4q_2, \quad 400 = 6q_1 + 4q_2.$$

Rearranging these equations for q_2 :

$$q_2 = -1.5q_1 + 50, \quad q_2 = -1.5q_1 + 75, \quad q_2 = -1.5q_1 + 100.$$

Answer: The total cost function is $TC(q_1, q_2) = 6q_1 + 4q_2$.

Contour map is shown below.



💡 Tip: Understanding Cost Optimization

A firm minimizes costs by choosing the production combination where the isocost line is **tangent** to the isoquant. This represents the lowest budget that still allows production of $Q = 300$.

✓ Solution 2: (c) Estimating the Optimal Production Point

To determine the most cost-efficient combination of raw materials that achieves $Q = 300$, we analyze the behavior of the total cost function.

To estimate the minimum cost required to produce $Q = 300$, we gradually move the isocost line $TC = a$ downward until it intersects the isoquant at exactly one point.

Looking at the graphical representation above, the optimal production point occurs at approximately:

$$(q_1, q_2) \approx (34, 38).$$

At this point, the firm achieves a production output of $Q = 300$ while incurring the lowest possible cost.

Answer: The estimated optimal production amounts are $q_1 \approx 34$ and $q_2 \approx 38$, which minimize costs while producing exactly $Q = 300$.

💡 Tip: Finding the Cost-Minimizing Production Point

A firm minimizes costs by choosing a production combination where the isocost line is tangent to the isoquant. Mathematically, this means that the slope of the isoquant must match the slope of the isocost line.

✓ Solution 2: (d) Exact Calculation of the Optimal Production Point

To determine the exact values of q_1 and q_2 that minimize costs while producing exactly $Q = 300$, we solve the system analytically.

Step 1: Compute the Isoquant Slope

From part (a), we found that the isoquant equation for $Q = 300$ is:

$$q_2 = \frac{1500}{q_1} - 0.2q_1.$$

Differentiating both sides with respect to q_1 :

$$\frac{dq_2}{dq_1} = \frac{-1500}{q_1^2} - 0.2.$$

Step 2: Solve for q_1 Using the Tangency Condition

Setting the derivative equal to the known isocost slope -1.5 :

$$\frac{-1500}{q_1^2} - 0.2 = -1.5.$$

Rearrange:

$$\frac{1500}{q_1^2} = 1.3 \quad \Rightarrow \quad q_1 = \sqrt{\frac{1500}{1.3}} \approx 33.97.$$

Step 3: Solve for q_2

Substituting $q_1 \approx 33.97$ into the isoquant equation:

$$q_2 = \frac{1500}{33.97} - 0.2(33.97) \quad \Rightarrow \quad q_2 \approx 37.37.$$

Answer: The exact optimal production amounts are:

$$(q_1, q_2) = (33.97, 37.37),$$

? Question 3

A firm produces two unrelated products with inverse demand functions:

$$\begin{cases} p_1 = 120 - 3q_1, \\ p_2 = 192 - 6q_2. \end{cases}$$

where p_1 and p_2 denote the unit prices for the products (in some unspecified monetary unit) and q_1 and q_2 the amounts demanded (in some unspecified unit as well). The production costs are given by the function:

$$C(q_1, q_2) = 2q_1^2 + 16q_1 + 6 + 2q_2^2.$$

- (a) Determine the domain of the cost function.
- (b) Make a contour plot for the cost function (*taking the entire context into account!*).
- (c) Indicate on the drawing you made in (b):
 - For which production amounts the cost is minimal.
 - For which production amounts the cost is maximal.

Explain clearly how you *use the contour plot from (b)* to arrive at these points.

- (d) Estimate the range of the cost function.

✓ Solution 3: (a) Determining the Domain of the Cost Function**Step 1: Identify Mathematical Constraints**

The cost function is given as:

$$C(q_1, q_2) = 2q_1^2 + 16q_1 + 6 + 2q_2^2.$$

Since this function is a sum of quadratic terms, it is well-defined for all real values of q_1 and q_2 . Thus, there are no mathematical restrictions.

Step 2: Determine the Practical Constraints

From the problem statement, the products follow the inverse demand functions:

$$\begin{cases} p_1 = 120 - 3q_1, \\ p_2 = 192 - 6q_2. \end{cases}$$

For the prices to remain non-negative:

$$p_1 = 120 - 3q_1 \geq 0 \quad \Rightarrow \quad q_1 \leq 40.$$

$$p_2 = 192 - 6q_2 \geq 0 \quad \Rightarrow \quad q_2 \leq 32.$$

Since negative production is not possible, we must also have:

$$q_1 \geq 0, \quad q_2 \geq 0.$$

Answer: The domain of the cost function is:

$$0 \leq q_1 \leq 40, \quad 0 \leq q_2 \leq 32.$$

✓ Solution 3: (b) Finding the Contour Lines of the Cost Function

Step 1: General Contour Equation

The given cost function is:

$$C(q_1, q_2) = 2q_1^2 + 16q_1 + 6 + 2q_2^2.$$

Setting $C(q_1, q_2) = a$ for an arbitrary level a , we solve:

$$2q_1^2 + 16q_1 + 6 + 2q_2^2 = a \Rightarrow 2q_1^2 + 2q_2^2 + 16q_1 = a - 6.$$

Divide by 2:

$$q_1^2 + q_2^2 + 8q_1 = \frac{a - 6}{2}.$$

Complete the square for q_1 :

$$(q_1 + 4)^2 + q_2^2 = \frac{a}{2} + 13.$$

This equation represents a circle centered at $(-4, 0)$ with radius:

$$\sqrt{\frac{a}{2} + 13}.$$

Step 2: Existence Conditions

For a contour line to exist, the radius must be real, meaning:

$$\frac{a}{2} + 13 \geq 0.$$

Solving for a :

$$a \geq -26.$$

For values of $a < -26$, the contour line is empty.

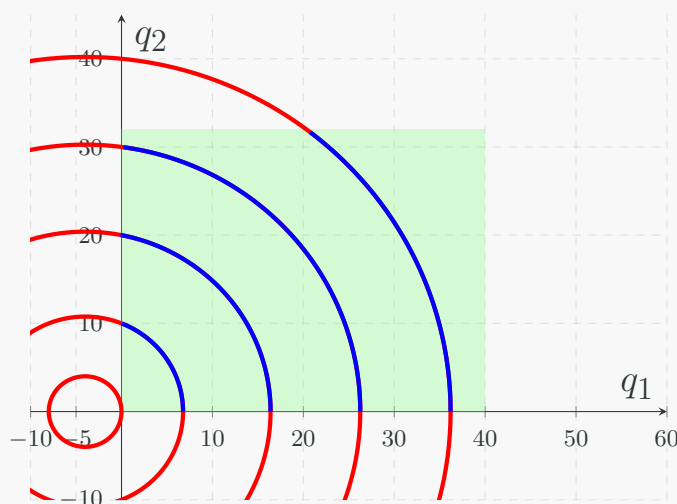
Step 3: Contextual Constraints

Although the mathematical contour is a full circle, only the portion within the domain $0 \leq q_1 \leq 40$, $0 \leq q_2 \leq 32$ is relevant. To illustrate this, we plot contour circles passing through some domain points:

$$(0, 0), (0, 10), (0, 20), (0, 30), (0, 40).$$

These correspond to cost levels from $C(0, 0) = 6$ all the way up to $C(0, 40) = 3206$.

Answer: The contour lines of the cost function are circles. Only the portions inside the domain are relevant.



✓ Solution 3: (c) Identifying the Minimum and Maximum Cost Points

To determine where the cost is minimized or maximized, we analyze the contour plot and identify the smallest and largest cost values within the valid domain.

Step 1: Identifying the Minimum Cost

The minimum cost occurs at the smallest contour circle that touches the green domain. From the contour plot, we see that the smallest red circle that intersects the domain touches at:

$$(q_1, q_2) = (0, 0).$$

Step 2: Identifying the Maximum Cost

The maximum cost occurs at the largest contour circle that still intersects the domain. This happens at the farthest point from the circle's center $(-4, 0)$, which is the upper-right corner of the green domain:

$$(q_1, q_2) = (40, 32).$$

Answer: The minimum cost occurs at $(q_1, q_2) = (0, 0)$. The maximum cost occurs at $(q_1, q_2) = (40, 32)$.

✓ Solution 3: (d) Estimating the Range of the Cost Function

To determine the range of the cost function, we analyze the minimum and maximum cost values found in part (c).

Step 1: Evaluating the Minimum Cost

From part (c), the minimum cost occurs at $(q_1, q_2) = (0, 0)$. Substituting this into the cost function:

$$C(0, 0) = 2(0)^2 + 16(0) + 6 + 2(0)^2 = 6.$$

Step 2: Evaluating the Maximum Cost

From part (c), the maximum cost occurs at $(q_1, q_2) = (40, 32)$. Substituting into the cost function:

$$C(40, 32) = 2(40)^2 + 16(40) + 6 + 2(32)^2 = 5886$$

Answer: The range of the cost function is $[6, 5886]$.

? Question 4

We consider the following continuous function of two variables:

$$f(x, y) = \frac{-4x^2 - 4y^2 + 112}{x - y}.$$

- (a) Make the sign chart for the function f . Make sure the figure is large enough because you have to complete it in question (b).
- (b) Determine the equation of the contour line of f passing through the point $(x, y) = (2, 4)$ and add this contour line to the figure you made in (a).

💡 Tip: Key Steps for Sign Charts

To determine where a function of two variables is positive, negative, or undefined, follow these three steps:

1. **Find the domain** – Check for any restrictions on x and y .
2. **Find where the function is zero** – Solve $f(x, y) = 0$ to identify boundary lines or curves.
3. **Find the sign in each region** – Pick test points in different regions to determine positivity or negativity.

✓ Solution 4: (a) Constructing the Sign Chart for $f(x, y)$ **Step 1: Find the domain**

The function is given by:

$$f(x, y) = \frac{-4x^2 - 4y^2 + 112}{x - y}.$$

The denominator must not be zero:

$$x - y \neq 0 \quad \Rightarrow \quad y \neq x.$$

Thus, the function is undefined along the line $y = x$, meaning this line must be excluded.

Since there are no square roots or logarithmic constraints, the function is otherwise defined for all real (x, y) .

Step 2: Find where the function is zero

The function is zero when the numerator equals zero:

$$-4x^2 - 4y^2 + 112 = 0.$$

Rearrange:

$$x^2 + y^2 = 28.$$

This equation represents a circle centered at $(0,0)$ with radius $\sqrt{28}$. Points on this circle correspond to $f(x, y) = 0$.

Step 3: Identify the sign in each region

The function is undefined along the line $y = x$, and it is zero along the circle $x^2 + y^2 = 28$.

These boundaries divide the plane into four distinct regions.

To determine the sign of $f(x, y)$ in each region, we select a test point in each area and compute $f(x, y)$.

- Test Point 1: $(-5, 5)$ (top-left region)

$$f(-5, 5) = \frac{-4(-5)^2 - 4(5)^2 + 112}{-5 - 5} = 8.8 > 0$$

Hence, the function is positive in this region.

- Test Point 2: $(5, -5)$ (bottom-right region)

$$f(5, -5) = \frac{-4(5)^2 - 4(-5)^2 + 112}{5 - (-5)} = -8.8 < 0$$

Hence, the function is negative in this region.

- Test Point 3: $(3, 0)$ (inside the circle, right side)

$$f(3, 0) = \frac{-4(3)^2 - 4(0)^2 + 112}{3 - 0} \approx 25.33 > 0$$

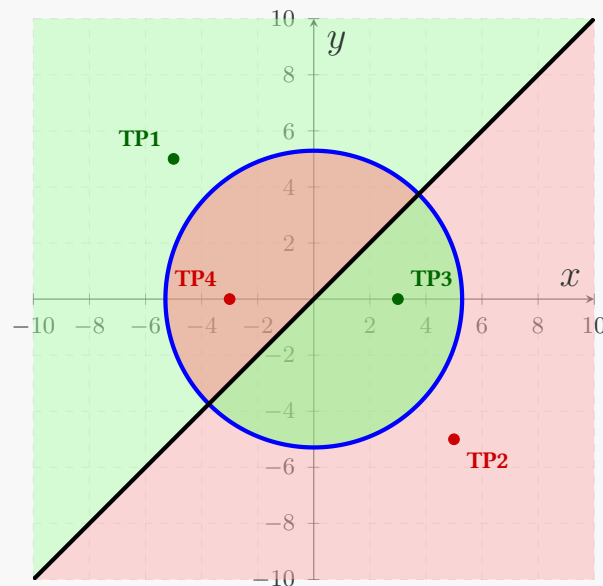
Hence, the function is positive in this region.

- Test Point 4: $(-3, 0)$ (inside the circle, left side)

$$f(-3, 0) = \frac{-4(-3)^2 - 4(0)^2 + 112}{-3 - 0} \approx -25.33 < 0$$

Hence, the function is negative in this region.

Answer: The green region represents where function is positive, the red region where it is negative, and the blue line where it is zero. The black line represents values that are not within the domain.



 Tip: Understanding Contour Lines

A **contour line** represents all points where the function attains the same value. To find it:

1. Set $f(x, y)$ equal to its value at the given point.
2. Solve for the equation that describes the level curve.
3. Identify any restrictions that modify the contour line within the domain.

 Solution 4: (b) Finding the Contour Line Passing Through (2, 4)**Step 1: Compute the Level Value**

We substitute (2, 4) into the function:

$$f(2, 4) = \frac{-4(2)^2 - 4(4)^2 + 112}{2 - 4} = -16$$

Step 2: Solve for the Contour Line

Setting $f(x, y) = -16$:

$$\frac{-4x^2 - 4y^2 + 112}{x - y} = -16 \quad \Rightarrow \quad -4x^2 - 4y^2 + 112 = -16(x - y).$$

Simplify:

$$x^2 + y^2 - 4x + 4y - 28 = 0.$$

Complete the square for x and y :

$$(x - 2)^2 + (y + 2)^2 = 36.$$

Thus, the contour line is a circle centered at (2, -2) with radius 6.

Step 3: Apply Domain Restrictions

As we have shown in part (a), the function is not defined along $y = x$. If the contour circle intersects this line, those points must be excluded.

Solving for intersections (by substituting $y = x$):

$$(x - 2)^2 + (x + 2)^2 = 36 \Rightarrow x = \pm\sqrt{14}.$$

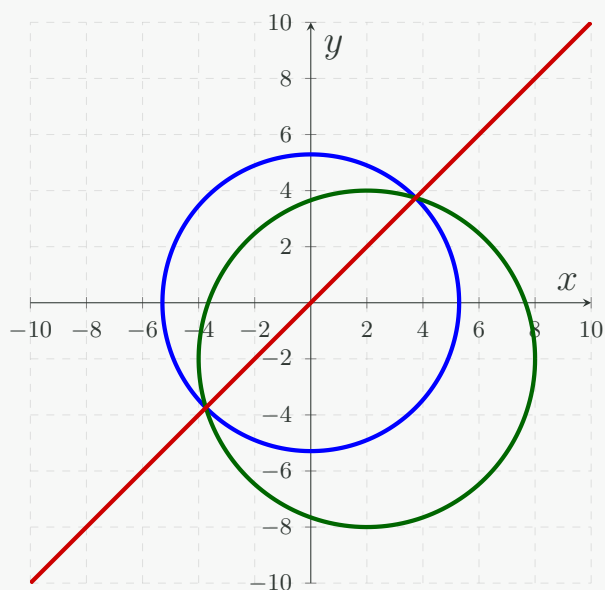
Since these points lie on $y = x$, they must be excluded.

Answer: The contour line that passes through $(2, 4)$ is given by the equation:

$$(x - 2)^2 + (y + 2)^2 = 36.$$

However, the points where this circle intersects the line $y = x$, specifically $(\pm\sqrt{14}, \pm\sqrt{14})$, are excluded.

The contour line is shown as the green circle on the graph below.



Chapter 2 - Linear functions and linear programming

Review 2 - Linear functions and linear programming

Chapter 3 - Partial Derivatives

Review 3 - Partial Derivatives

Chapter 4 - Optimization

Review 4 - Optimization

Chapter 5 - Sequences and Series

Review 5 - Sequences and Series

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